

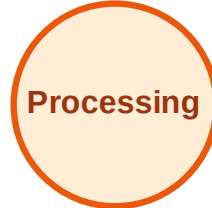
BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing



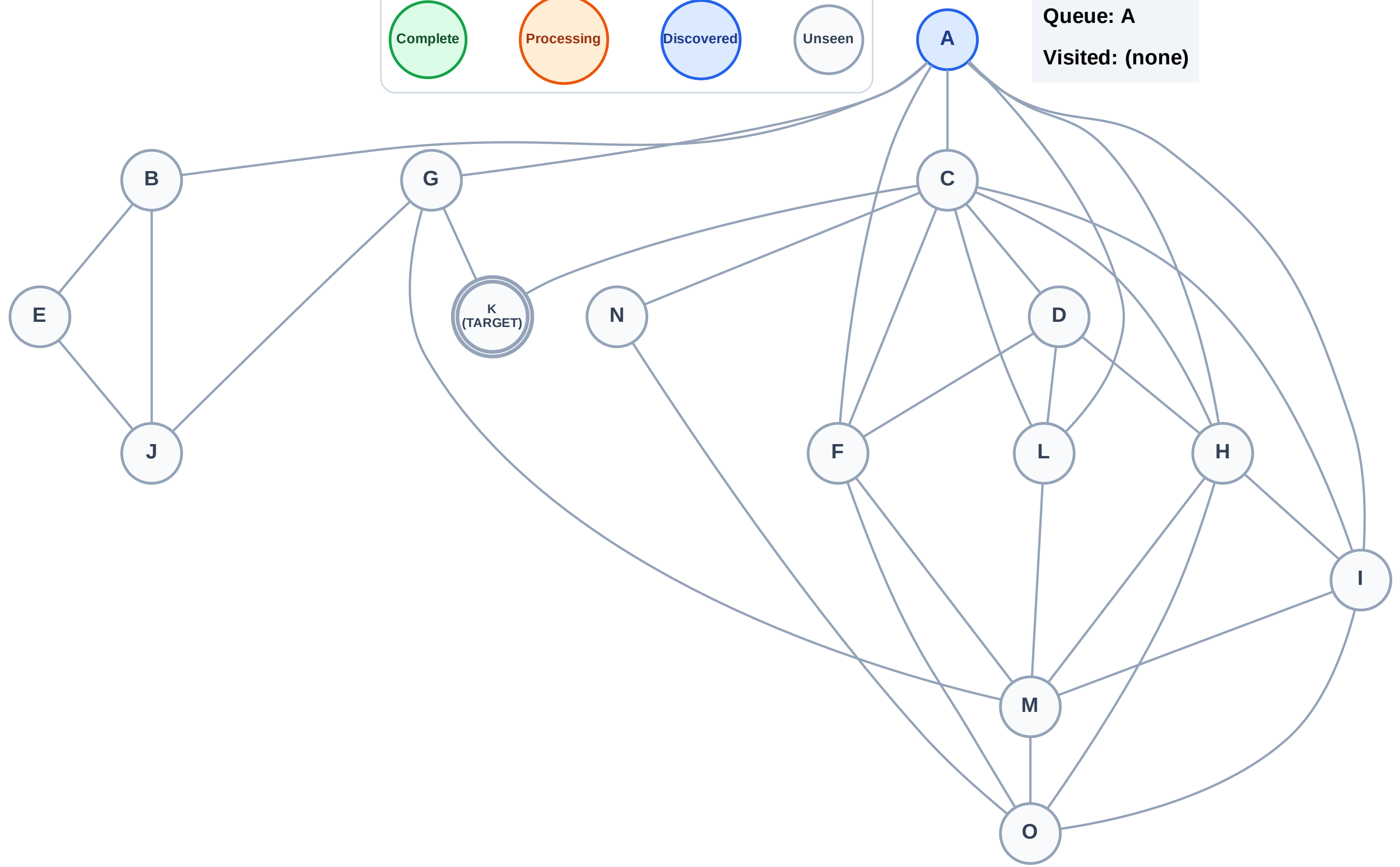
Discovered



Unseen

Queue: A

Visited: (none)



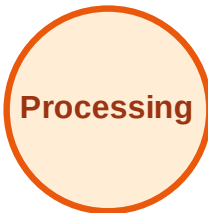
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



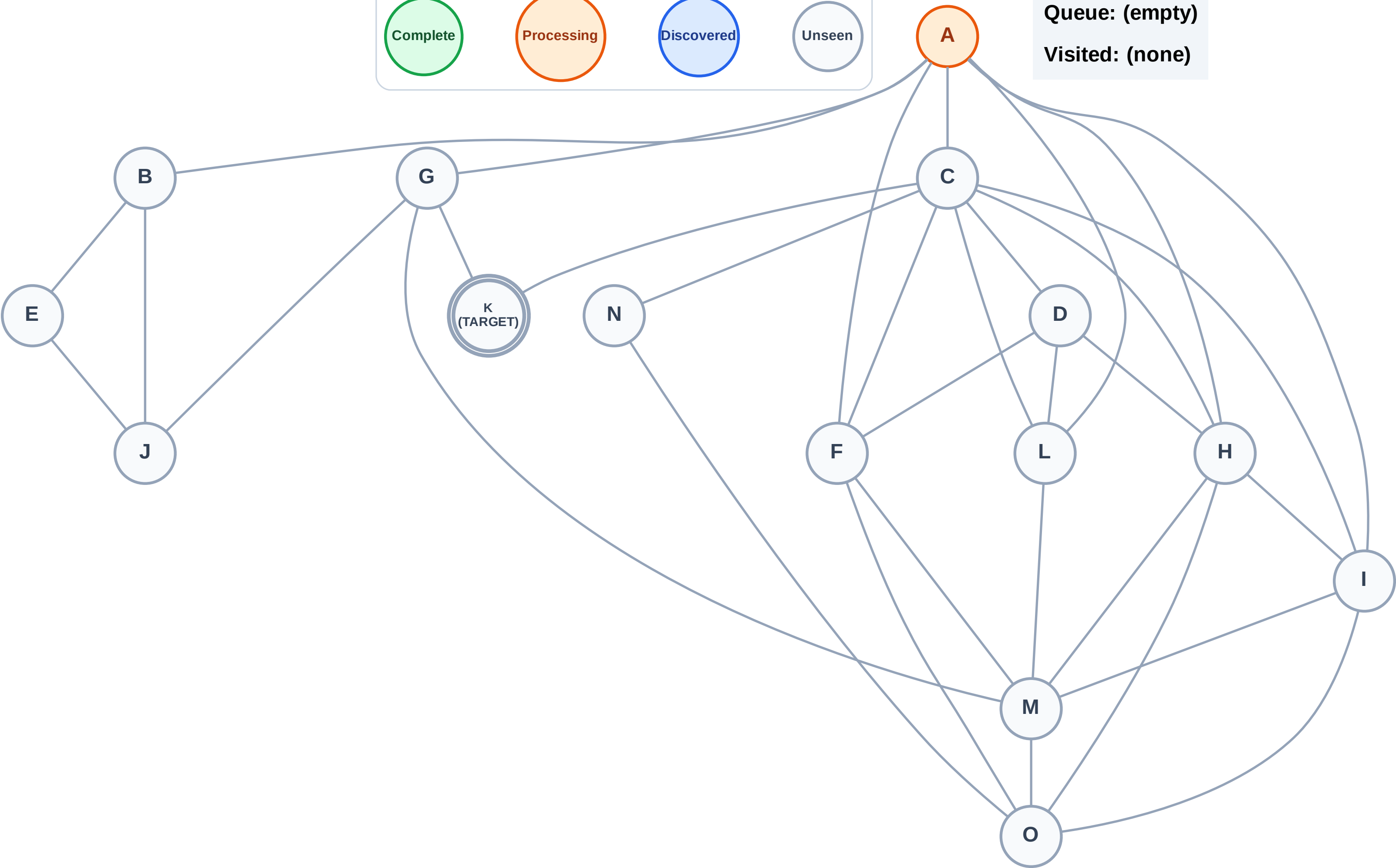
Discovered



Unseen

Queue: (empty)

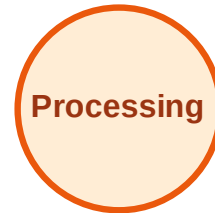
Visited: (none)



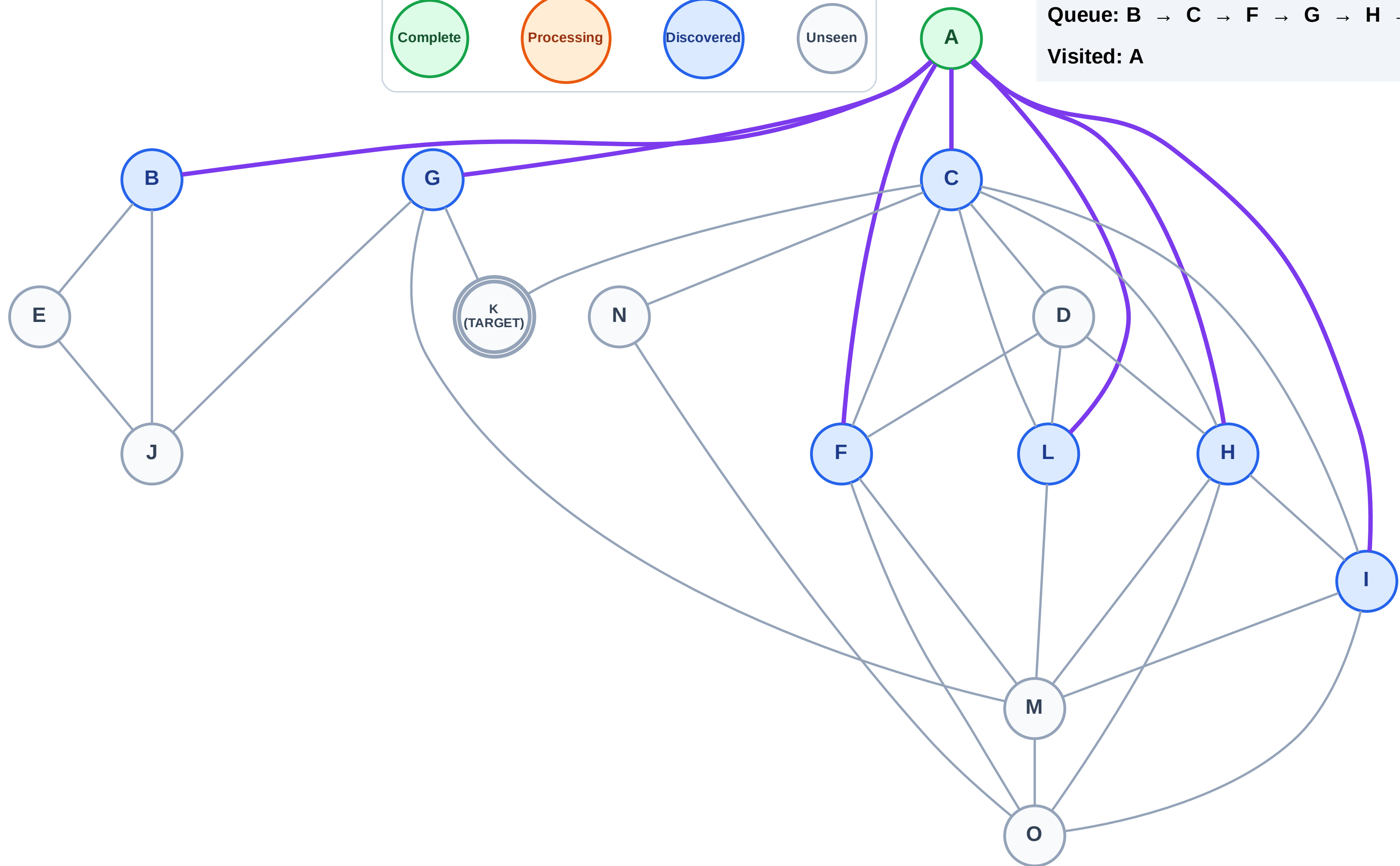
BFS Traversal

Step 3: Discover B, C, F, G, H, I, L from A

Legend



Queue: B → C → F → G → H → I → L
Visited: A



BFS Traversal

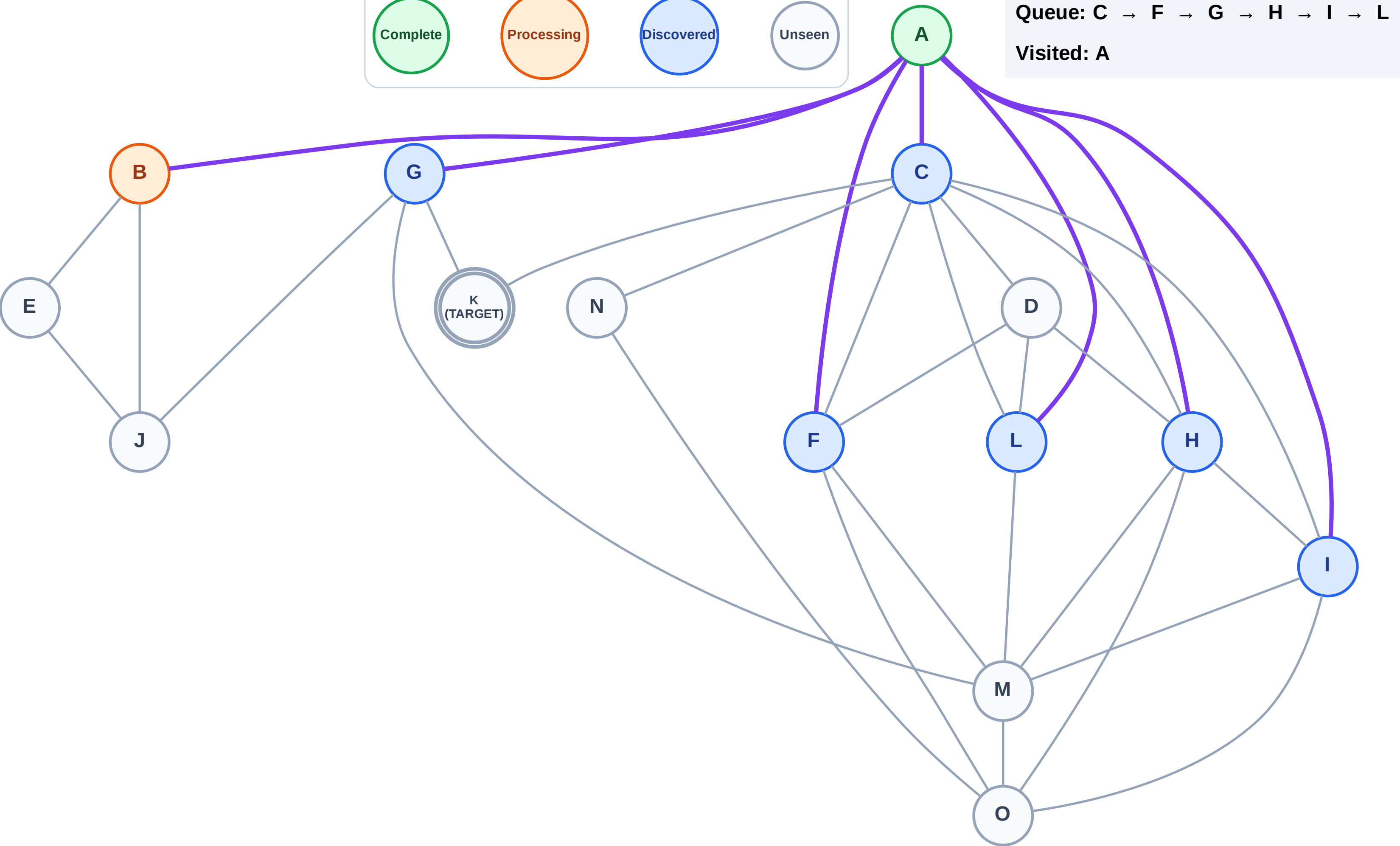
Step 4: Process node B

Legend



Queue: C → F → G → H → I → L

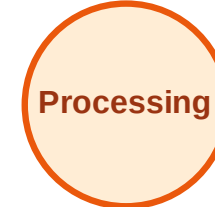
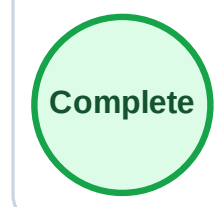
Visited: A



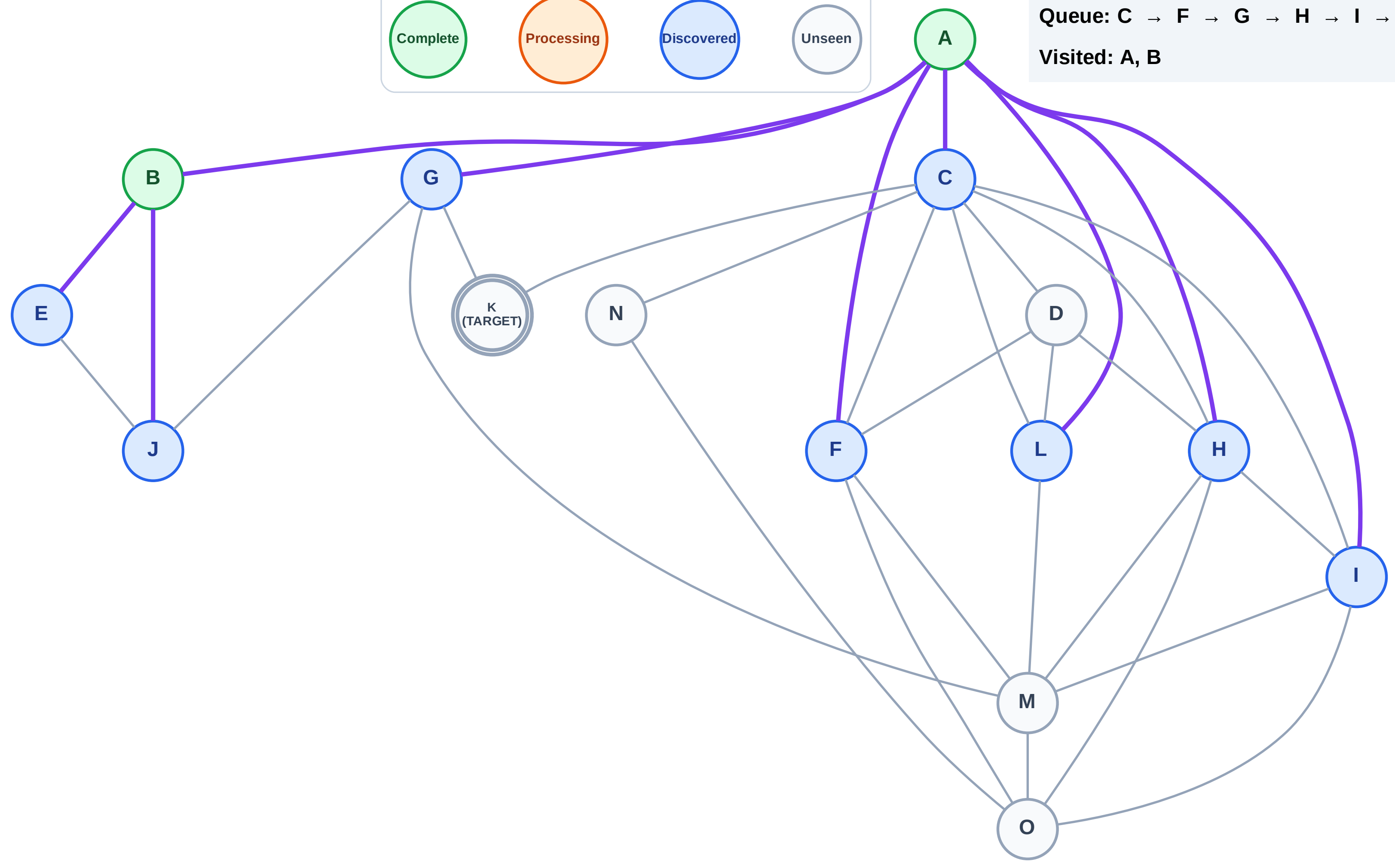
BFS Traversal

Step 5: Discover E, J from B

Legend



Queue: C → F → G → H → I → L → E → J
Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

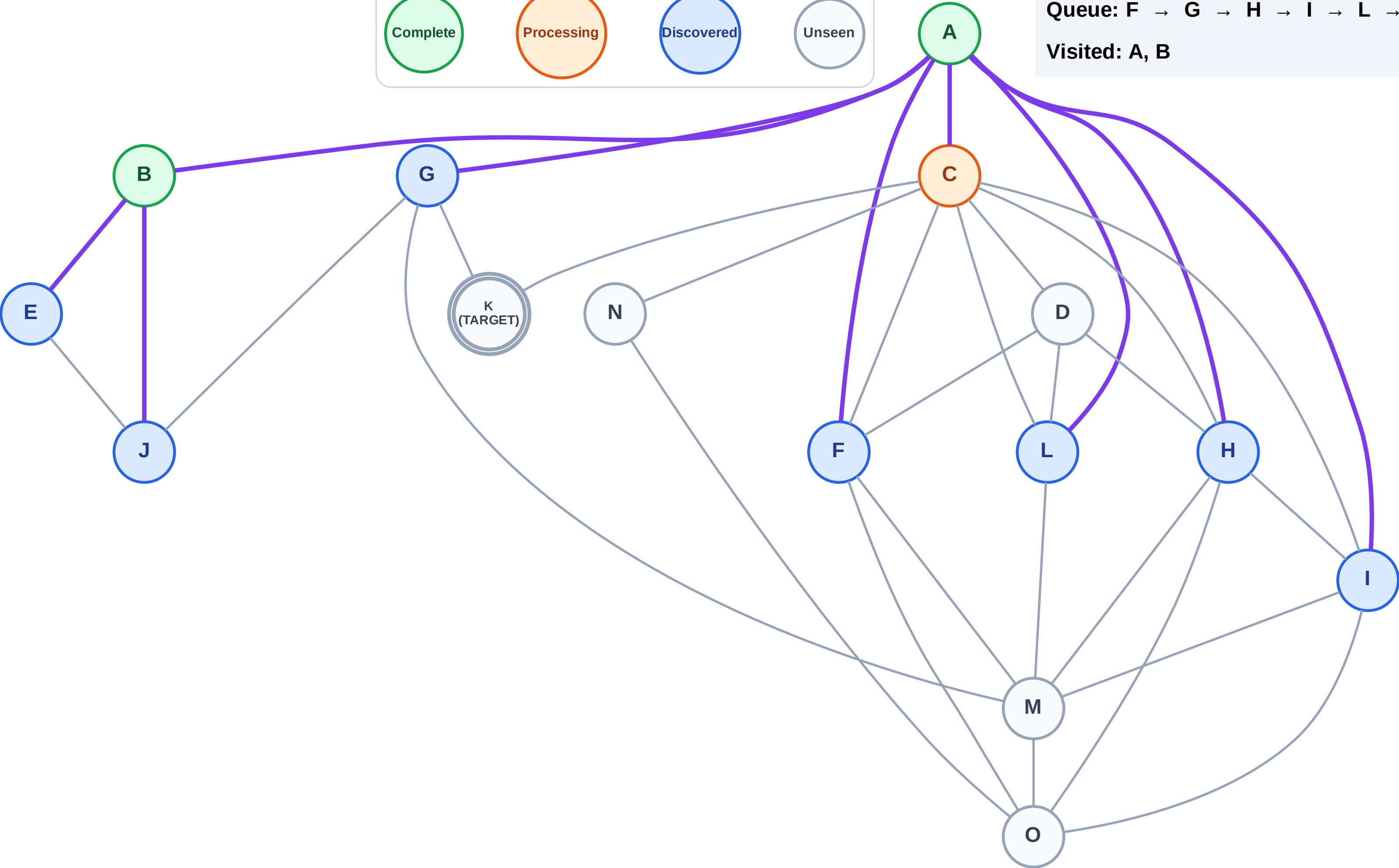
Processing

Discovered

Unseen

Queue: F → G → H → I → L → E → J

Visited: A, B



BFS Traversal

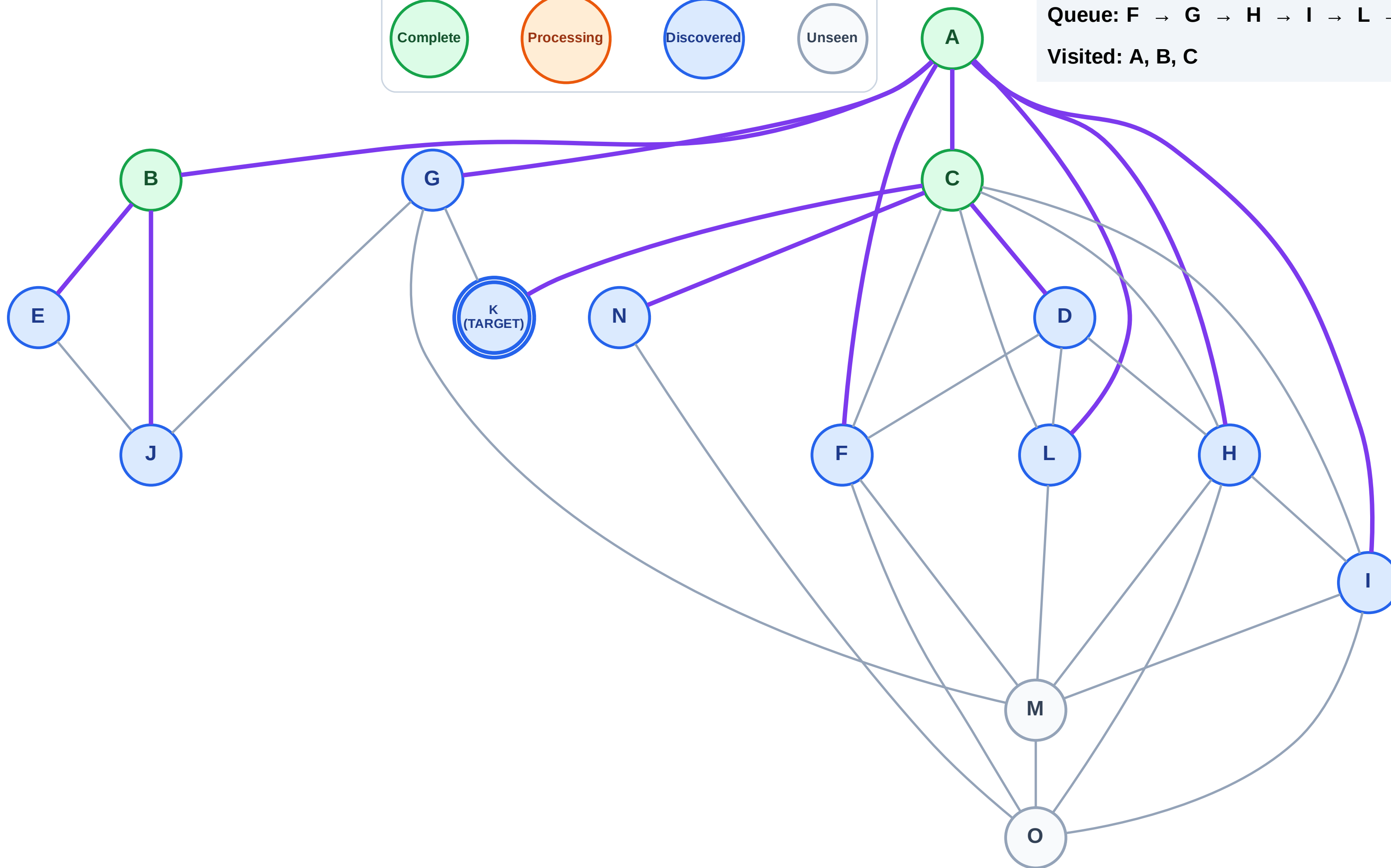
Step 7: Discover D, K, N from C

Legend



Queue: F → G → H → I → L → E → J → D → K → N

Visited: A, B, C



BFS Traversal

Step 8: Process node F

Legend

Complete

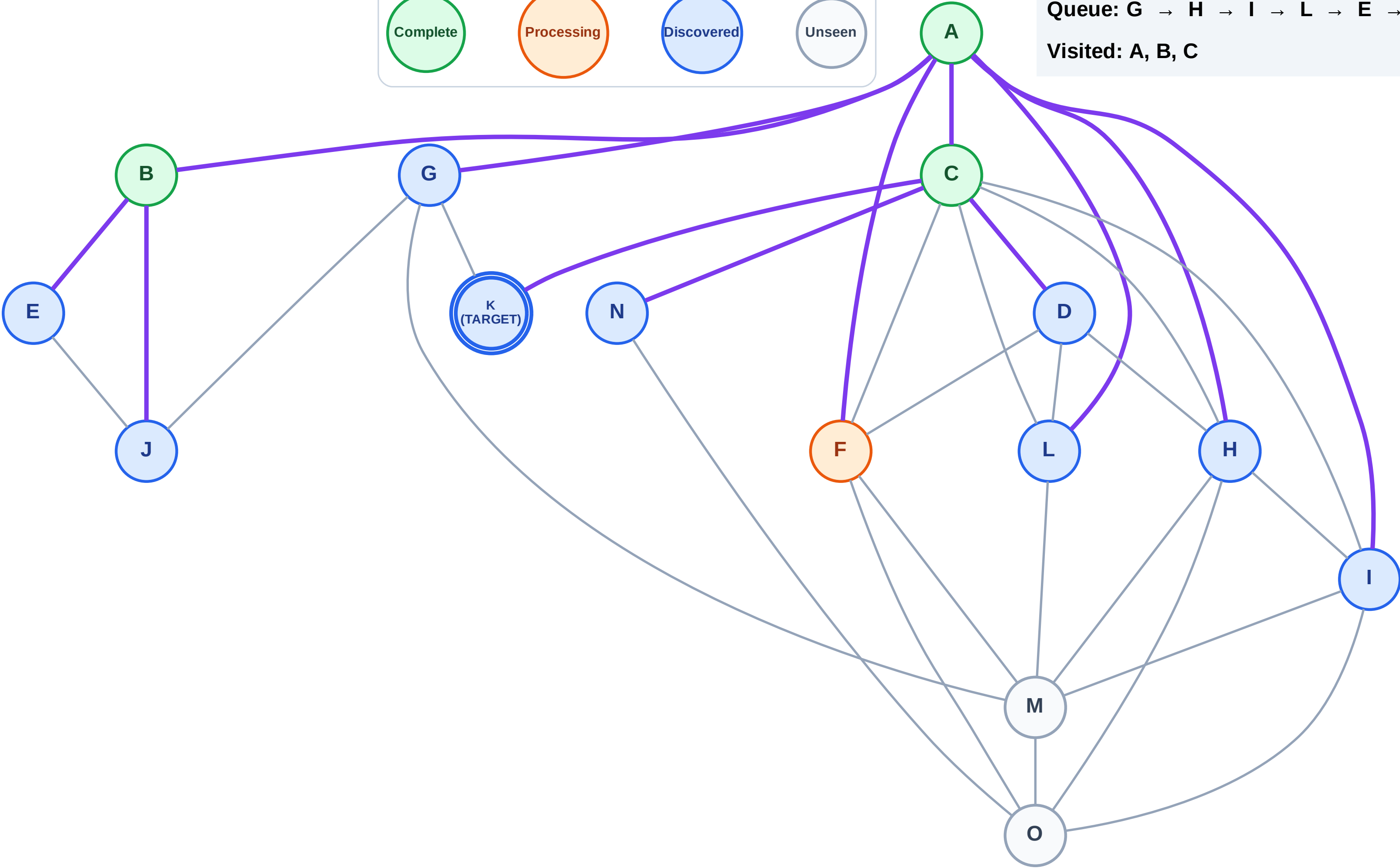
Processing

Discovered

Unseen

Queue: G → H → I → L → E → J → D → K → N

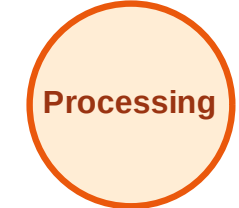
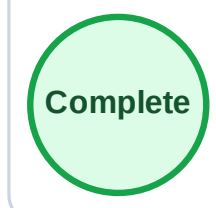
Visited: A, B, C



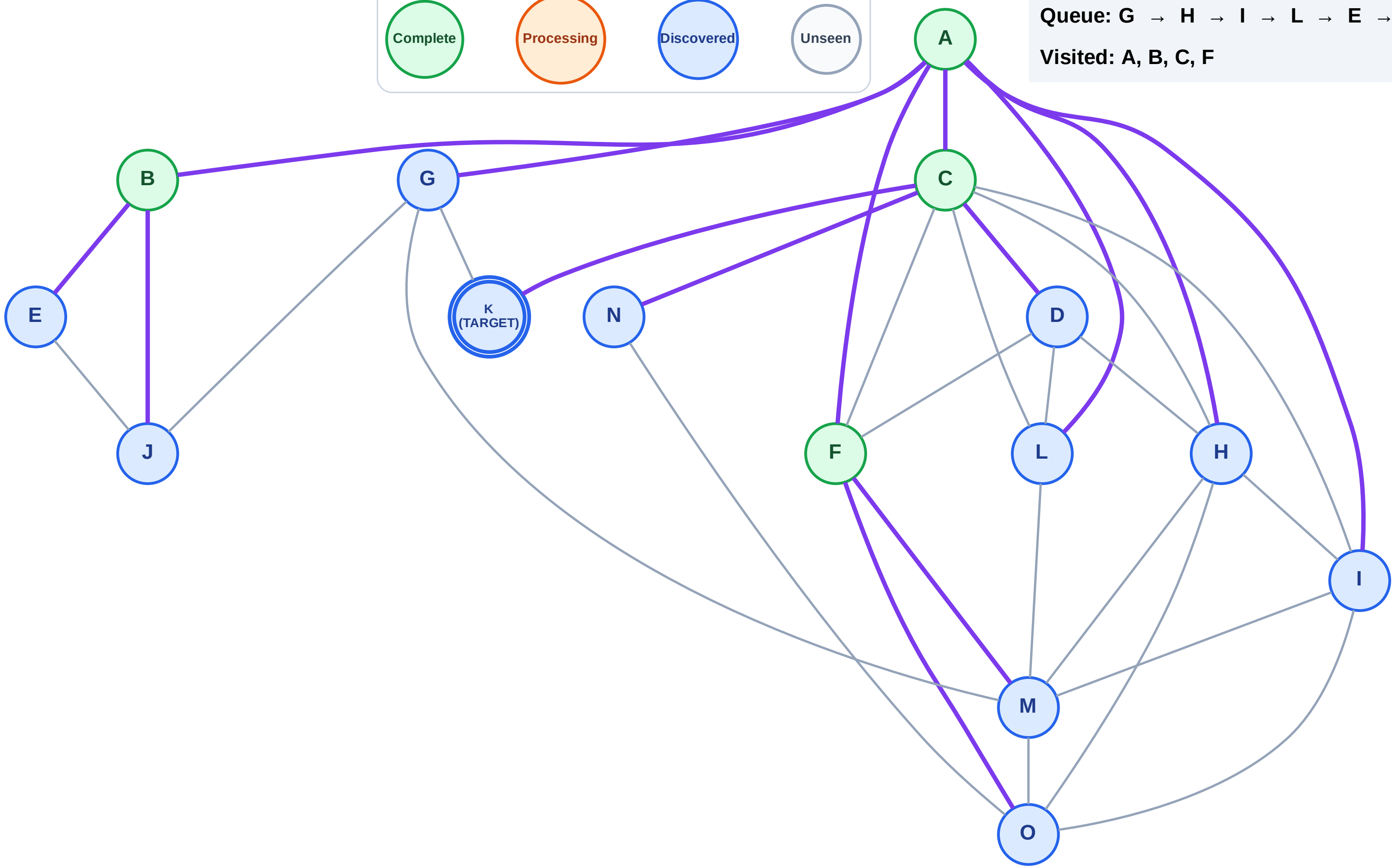
BFS Traversal

Step 9: Discover M, O from F

Legend



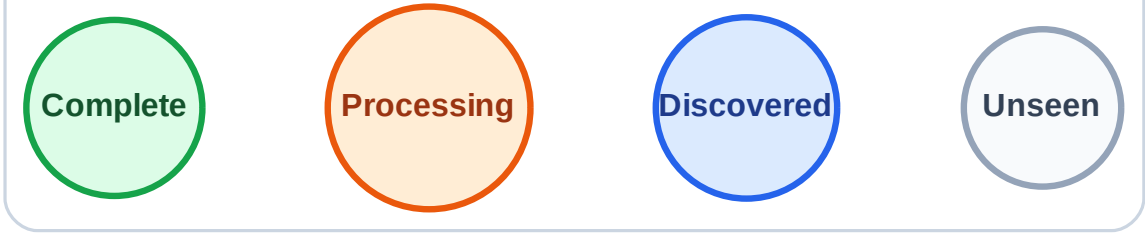
Queue: G → H → I → L → E → J → D → K → N → M → O
Visited: A, B, C, F



BFS Traversal

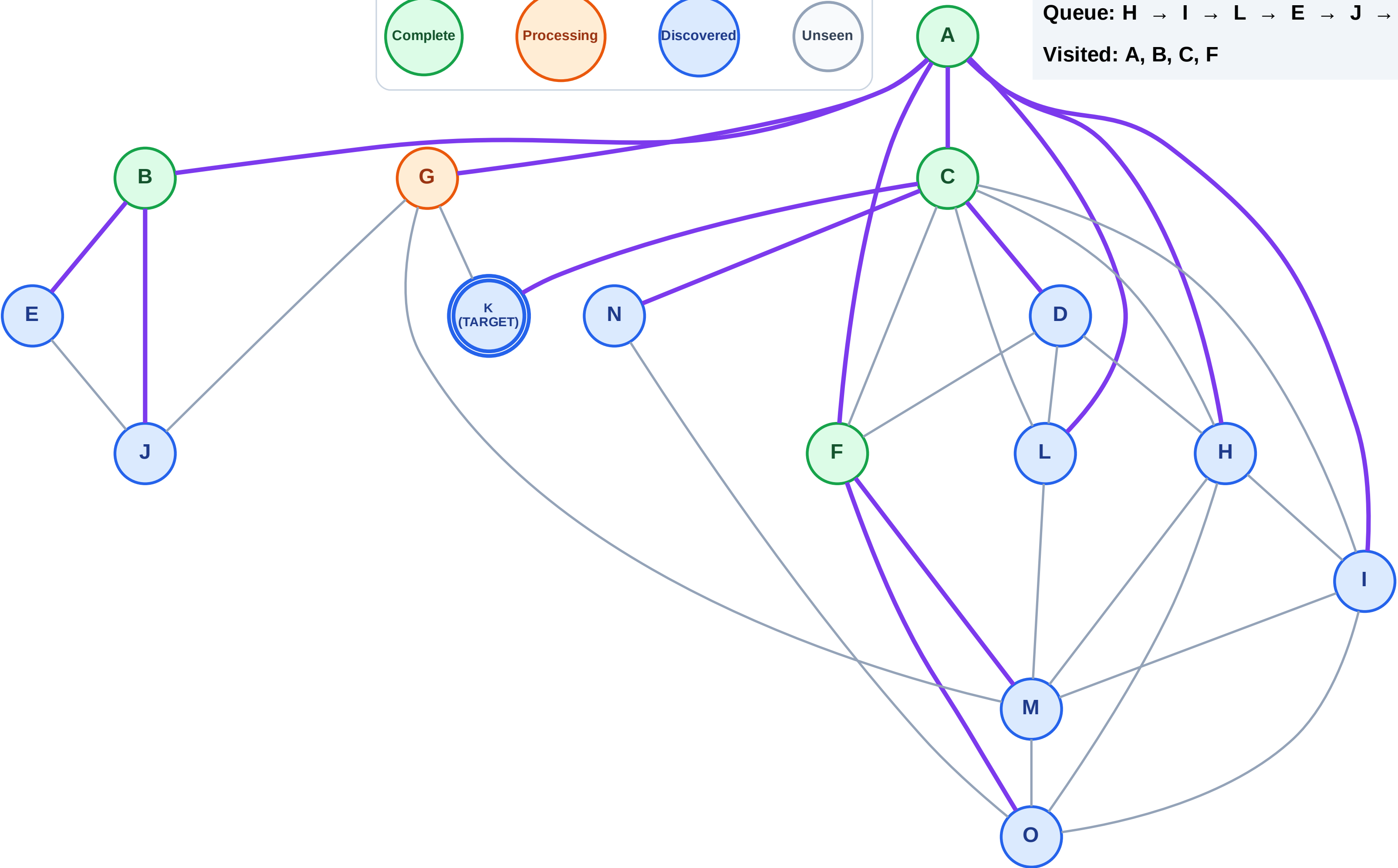
Step 10: Process node G

Legend



Queue: H → I → L → E → J → D → K → N → M → O

Visited: A, B, C, F



BFS Traversal

Step 11: No new neighbors from G

Legend



Complete



Processing



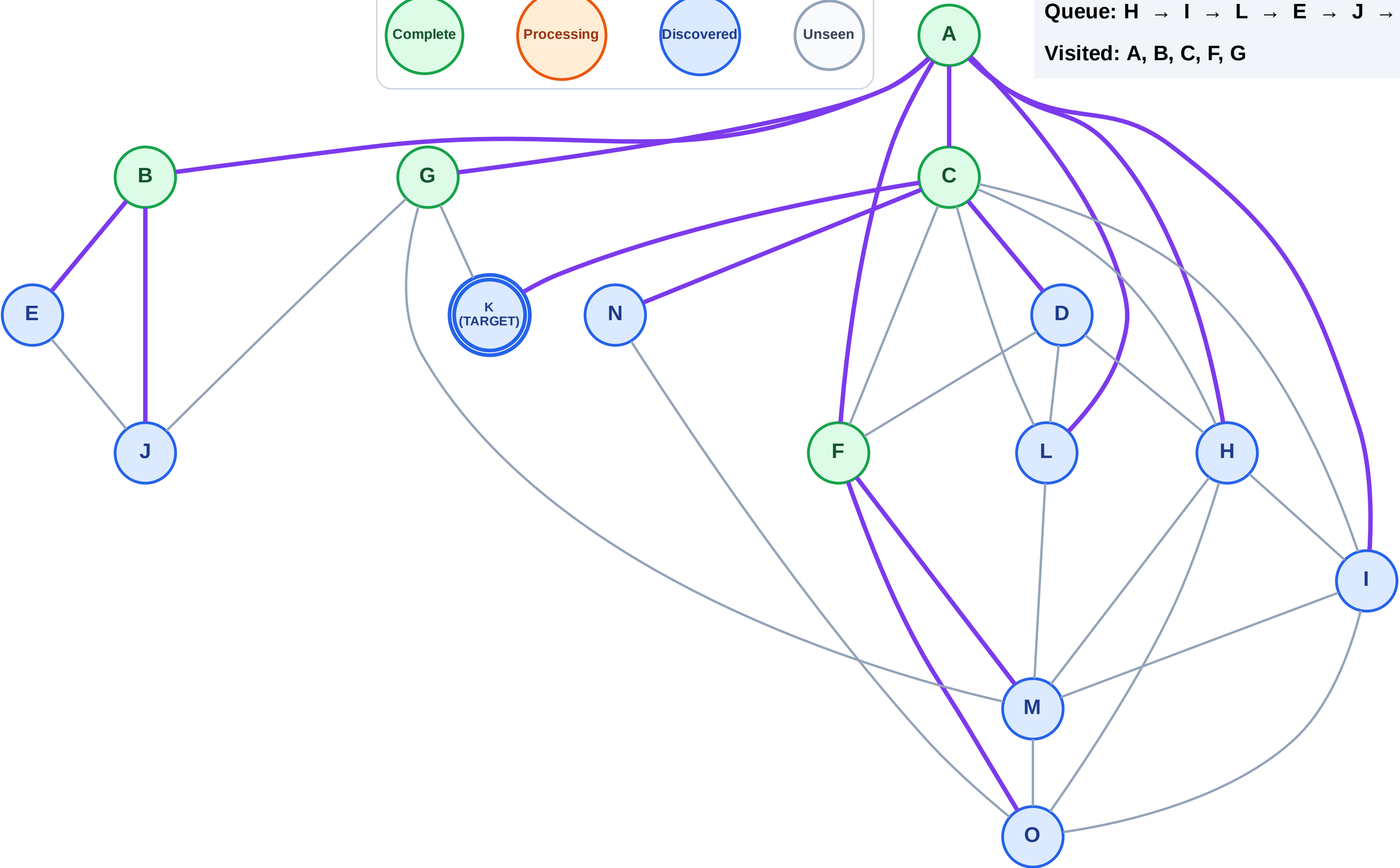
Discovered



Unseen

Queue: H → I → L → E → J → D → K → N → M → O

Visited: A, B, C, F, G



BFS Traversal

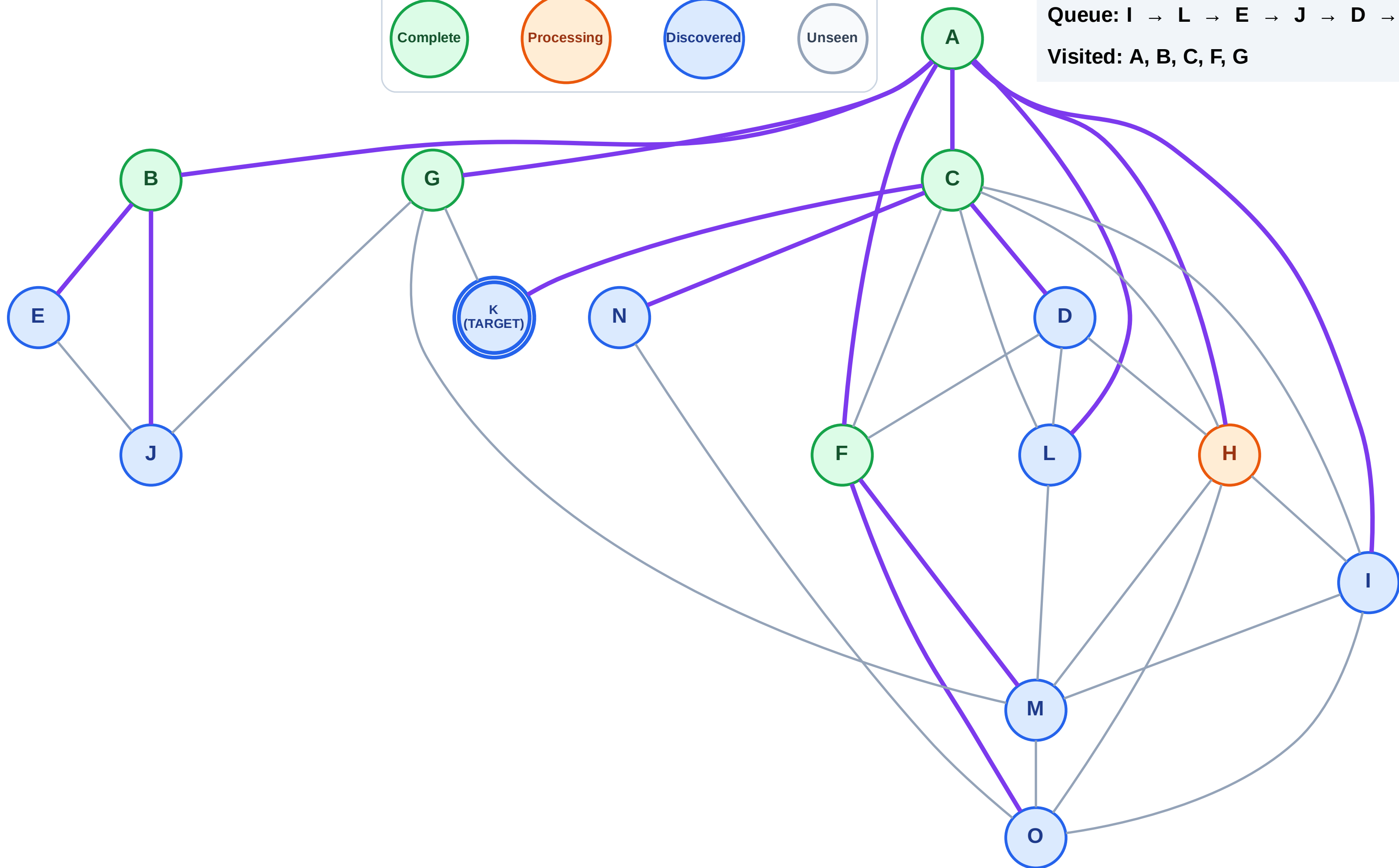
Step 12: Process node H

Legend



Queue: I → L → E → J → D → K → N → M → O

Visited: A, B, C, F, G



BFS Traversal

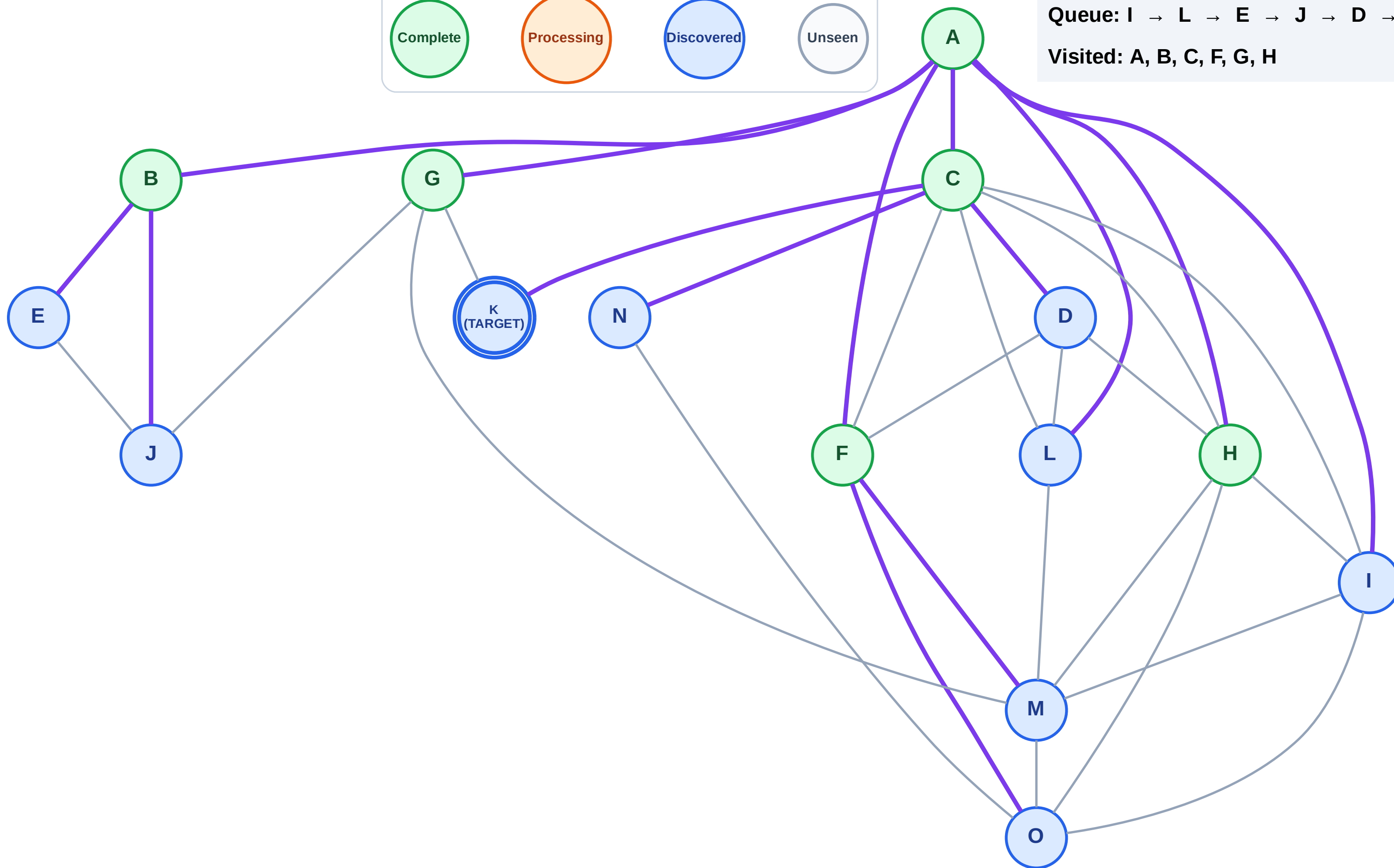
Step 13: No new neighbors from H

Legend



Queue: I → L → E → J → D → K → N → M → O

Visited: A, B, C, F, G, H



BFS Traversal

Step 14: Process node I

Legend

Complete

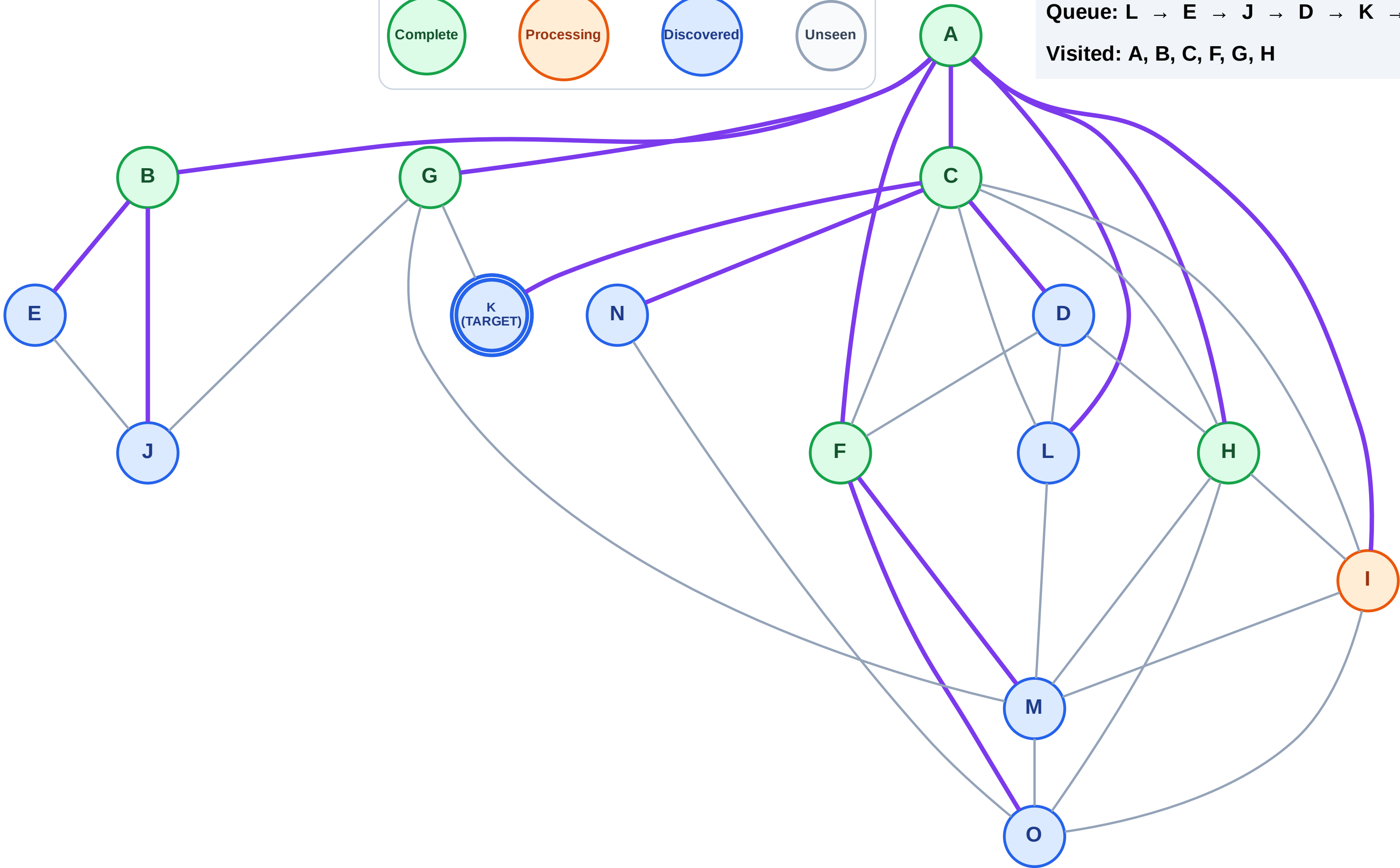
Processing

Discovered

Unseen

Queue: L → E → J → D → K → N → M → O

Visited: A, B, C, F, G, H



BFS Traversal

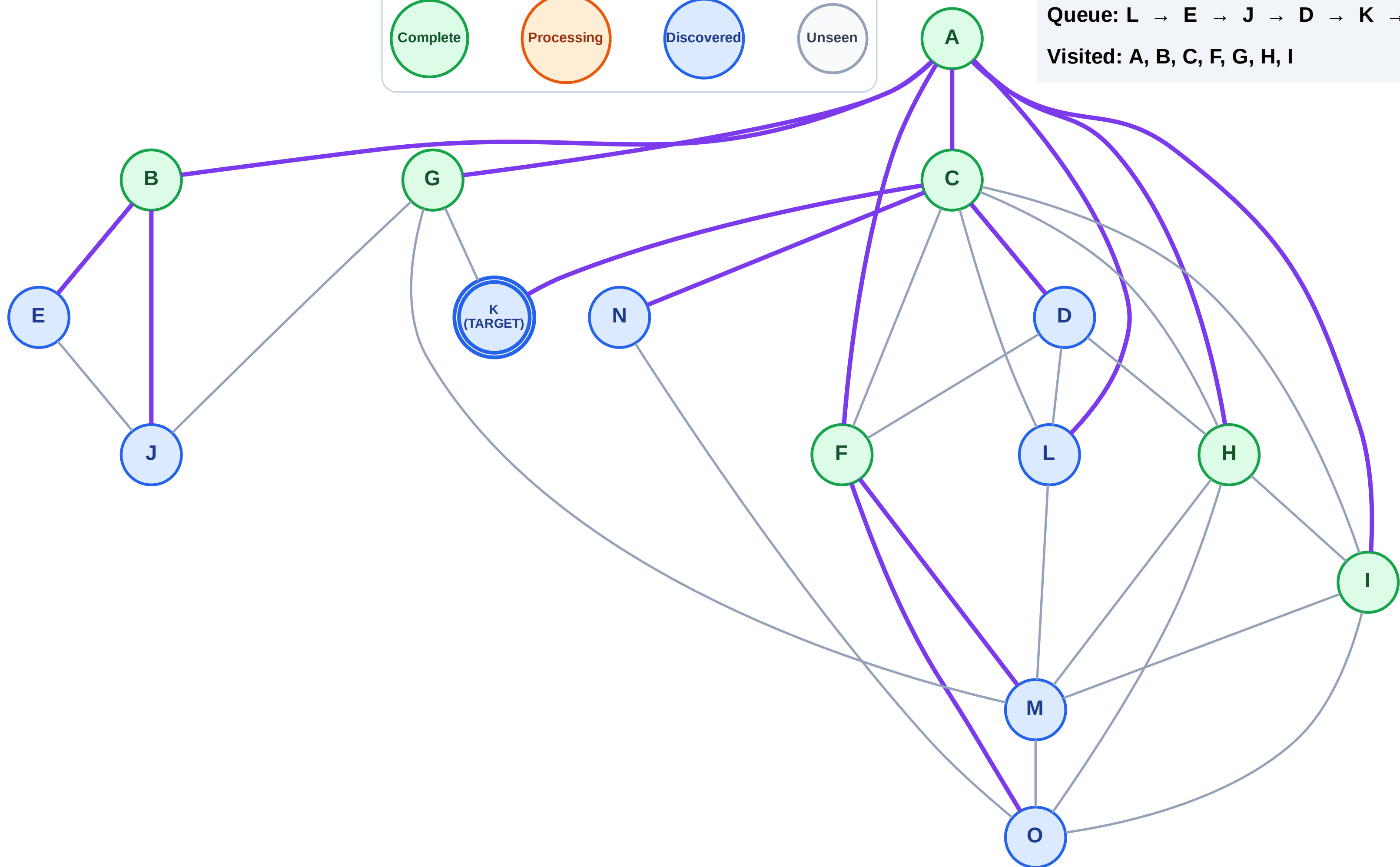
Step 15: No new neighbors from I

Legend



Queue: L → E → J → D → K → N → M → O

Visited: A, B, C, F, G, H, I



BFS Traversal

Step 16: Process node L

Legend

Complete

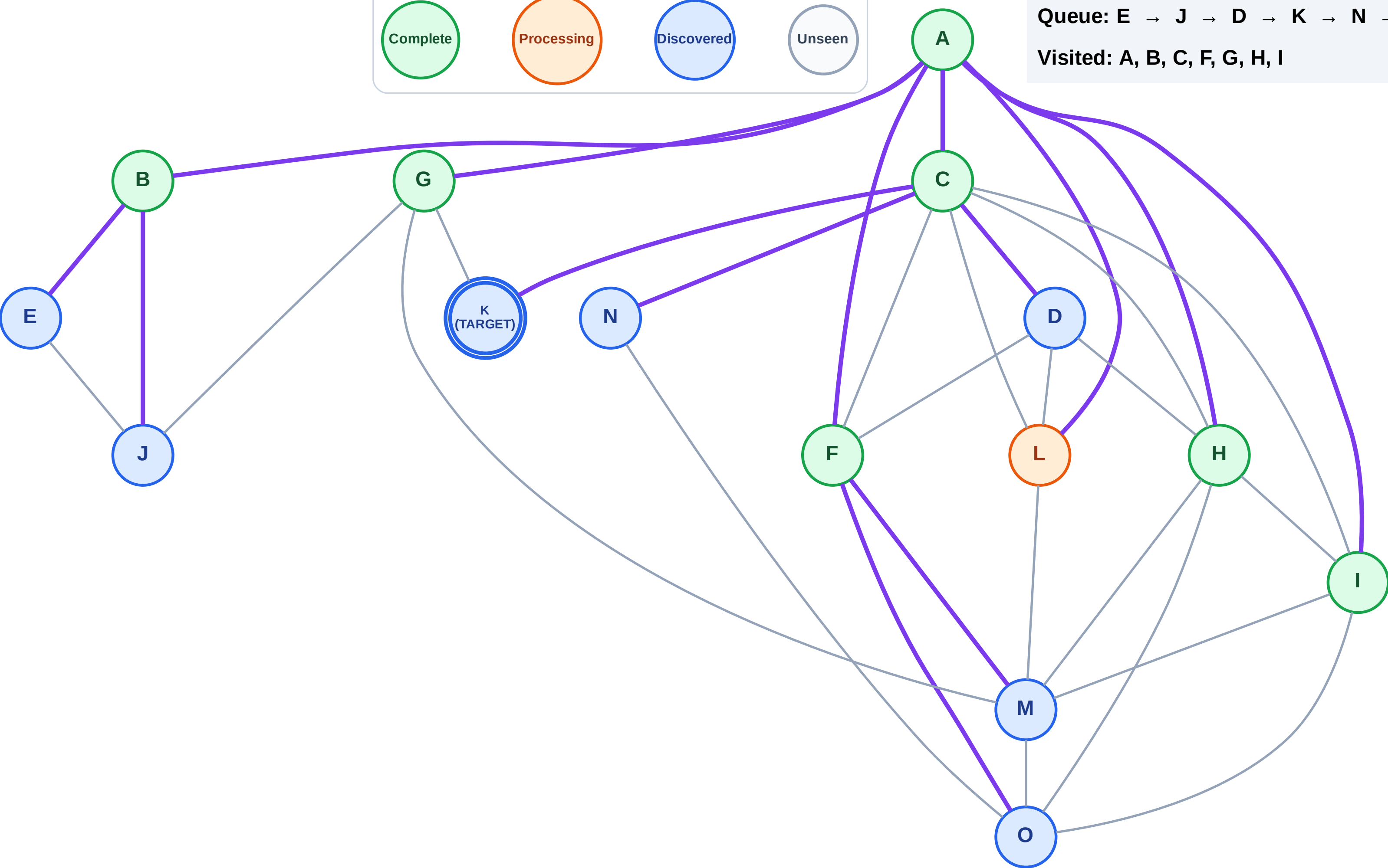
Processing

Discovered

Unseen

Queue: E → J → D → K → N → M → O

Visited: A, B, C, F, G, H, I



BFS Traversal

Step 17: No new neighbors from L

Legend

Complete

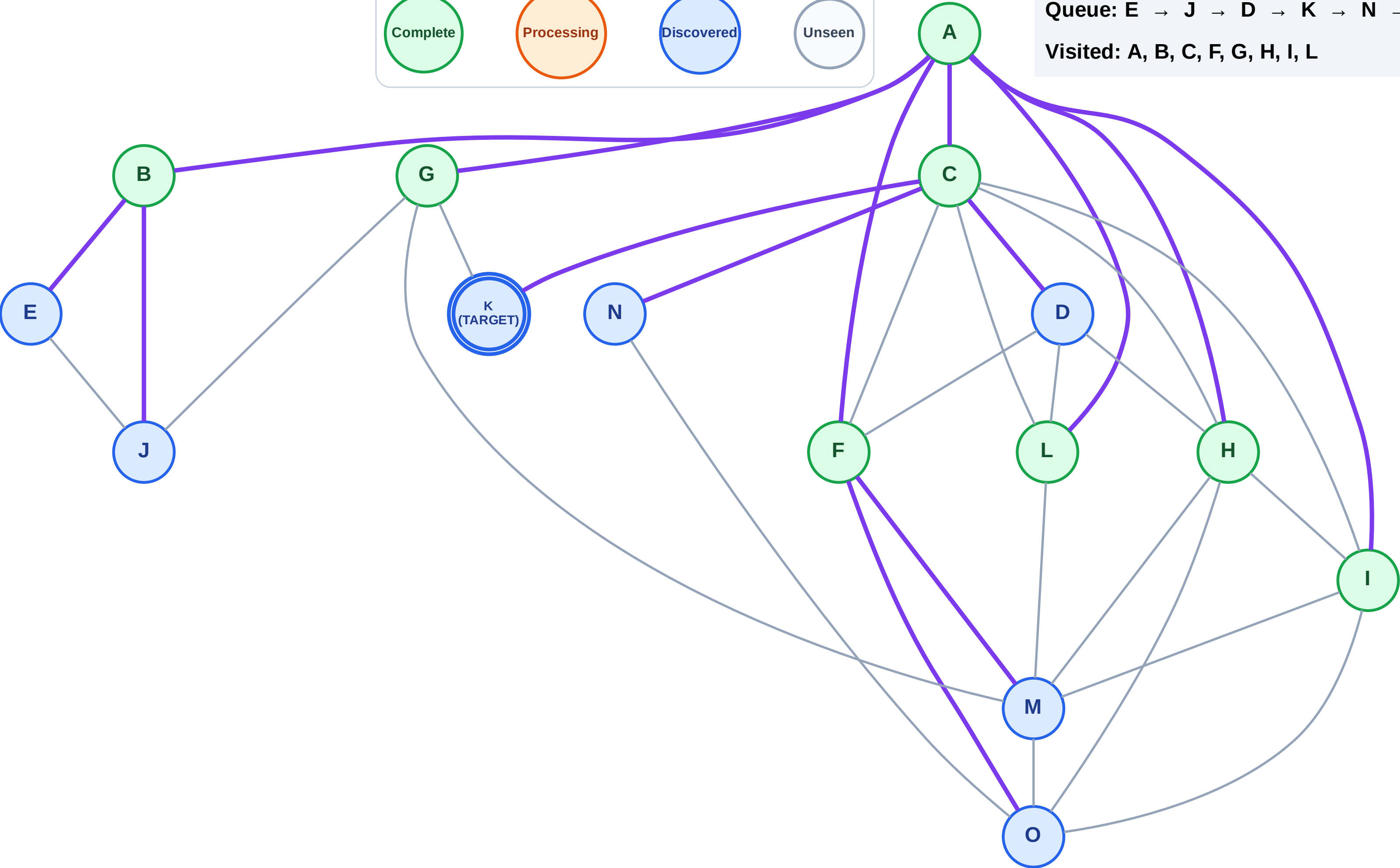
Processing

Discovered

Unseen

Queue: E → J → D → K → N → M → O

Visited: A, B, C, F, G, H, I, L



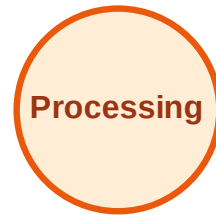
BFS Traversal

Step 18: Process node E

Legend



Complete



Processing



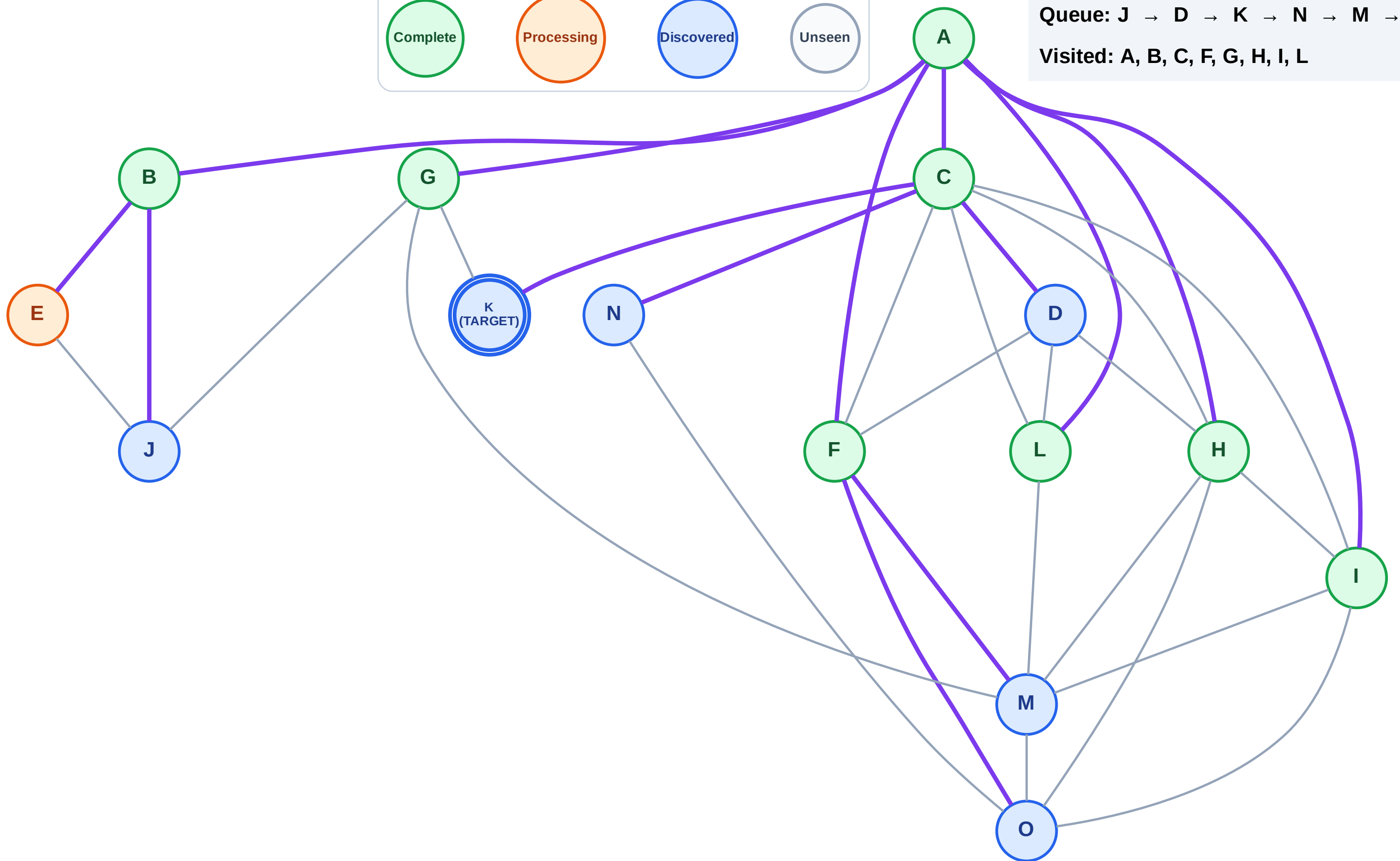
Discovered



Unseen

Queue: J → D → K → N → M → O

Visited: A, B, C, F, G, H, I, L



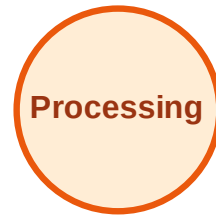
BFS Traversal

Step 19: No new neighbors from E

Legend



Complete



Processing



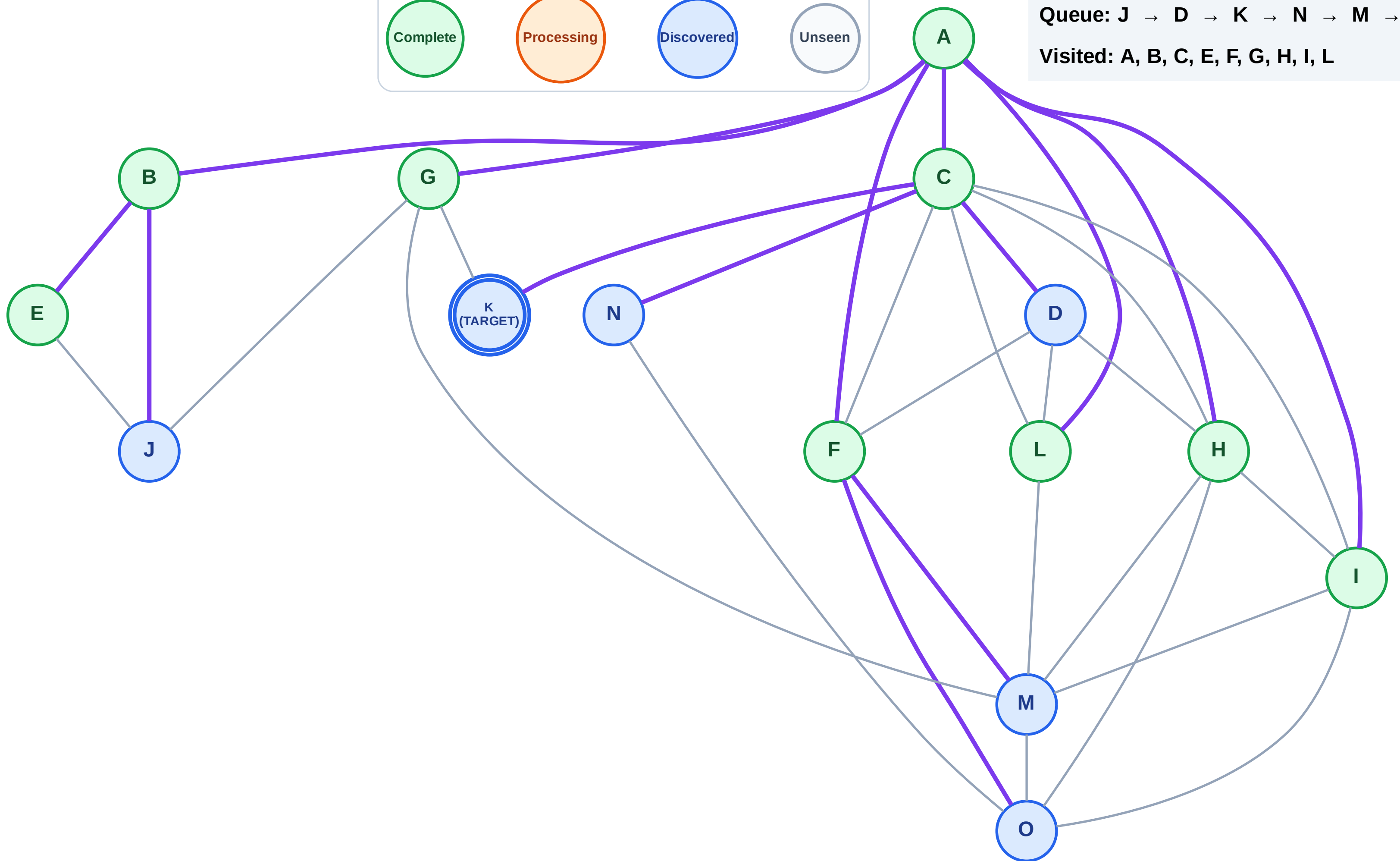
Discovered



Unseen

Queue: J → D → K → N → M → O

Visited: A, B, C, E, F, G, H, I, L



BFS Traversal

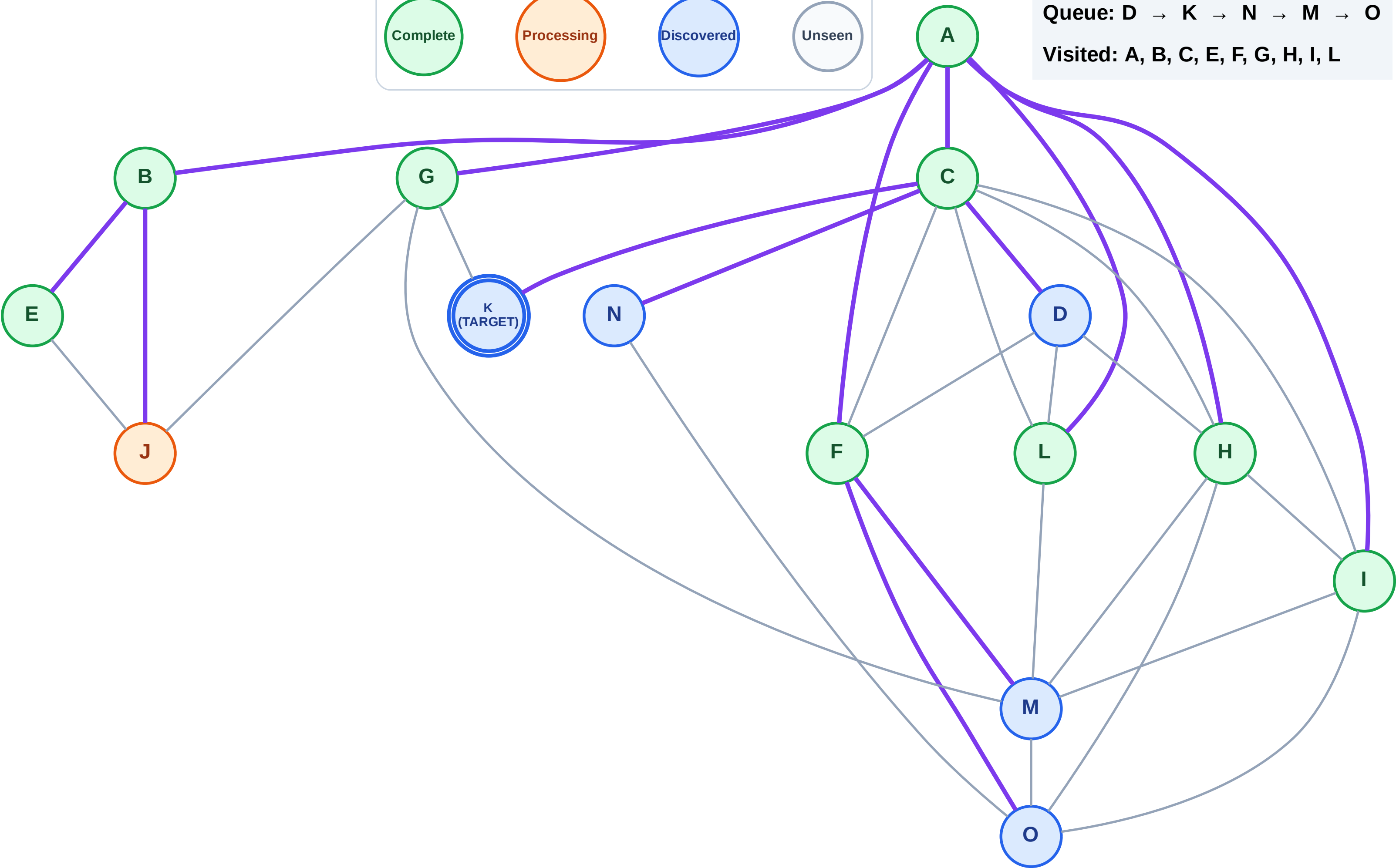
Step 20: Process node J

Legend



Queue: D → K → N → M → O

Visited: A, B, C, E, F, G, H, I, L



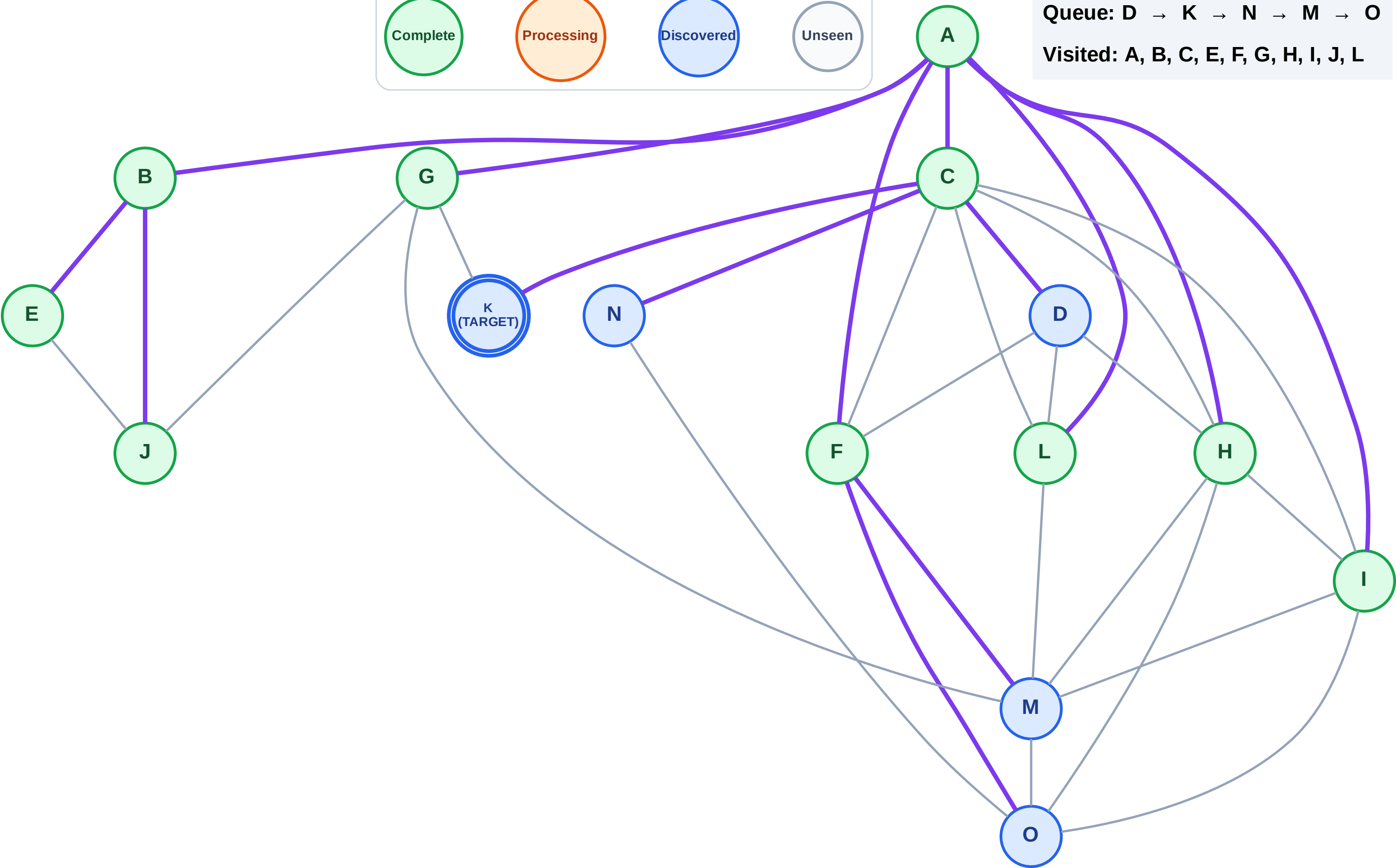
BFS Traversal

Step 21: No new neighbors from J

Legend



Queue: D → K → N → M → O
Visited: A, B, C, E, F, G, H, I, J, L



BFS Traversal

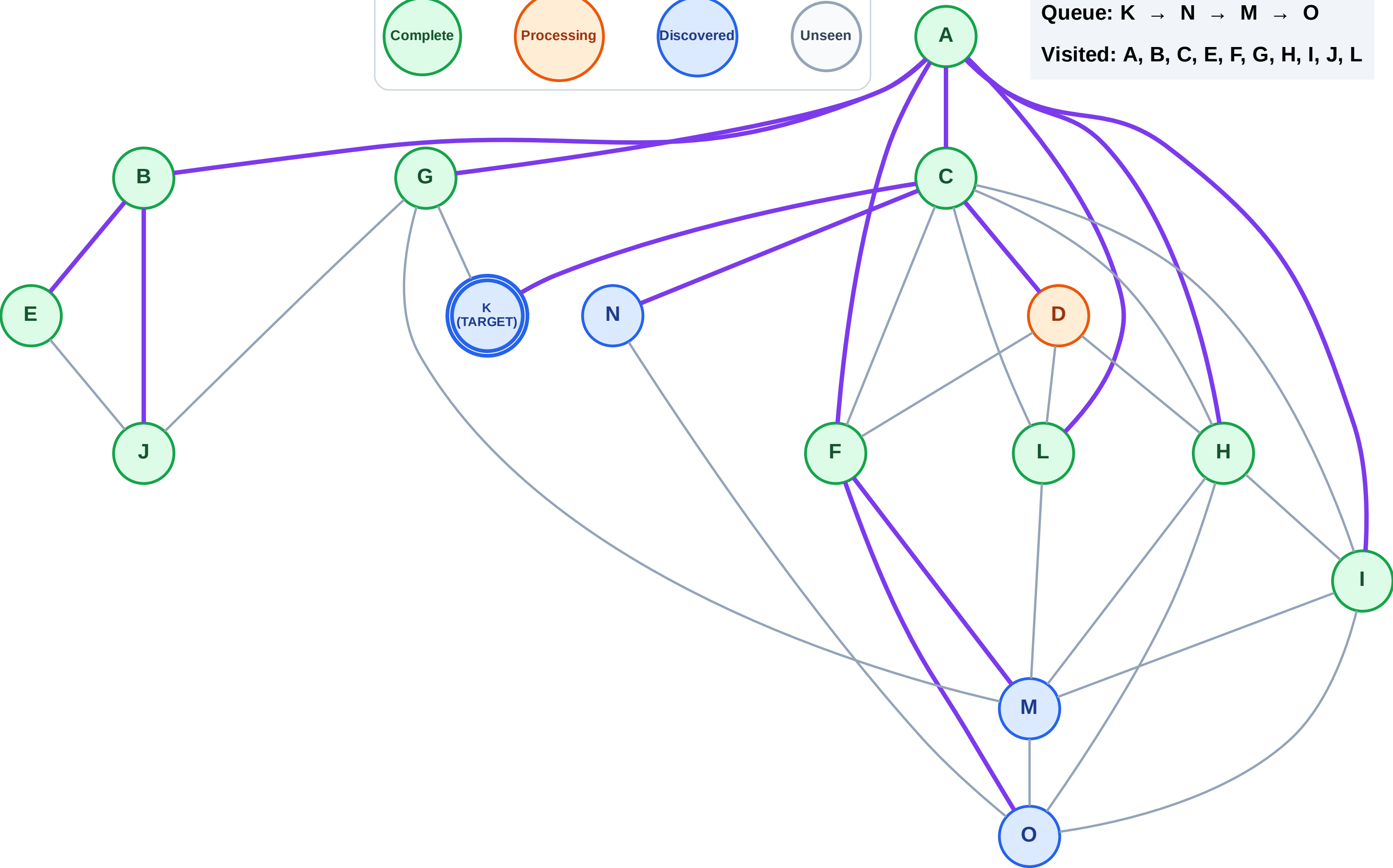
Step 22: Process node D

Legend



Queue: K → N → M → O

Visited: A, B, C, E, F, G, H, I, J, L



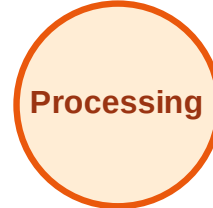
BFS Traversal

Step 23: No new neighbors from D

Legend



Complete



Processing



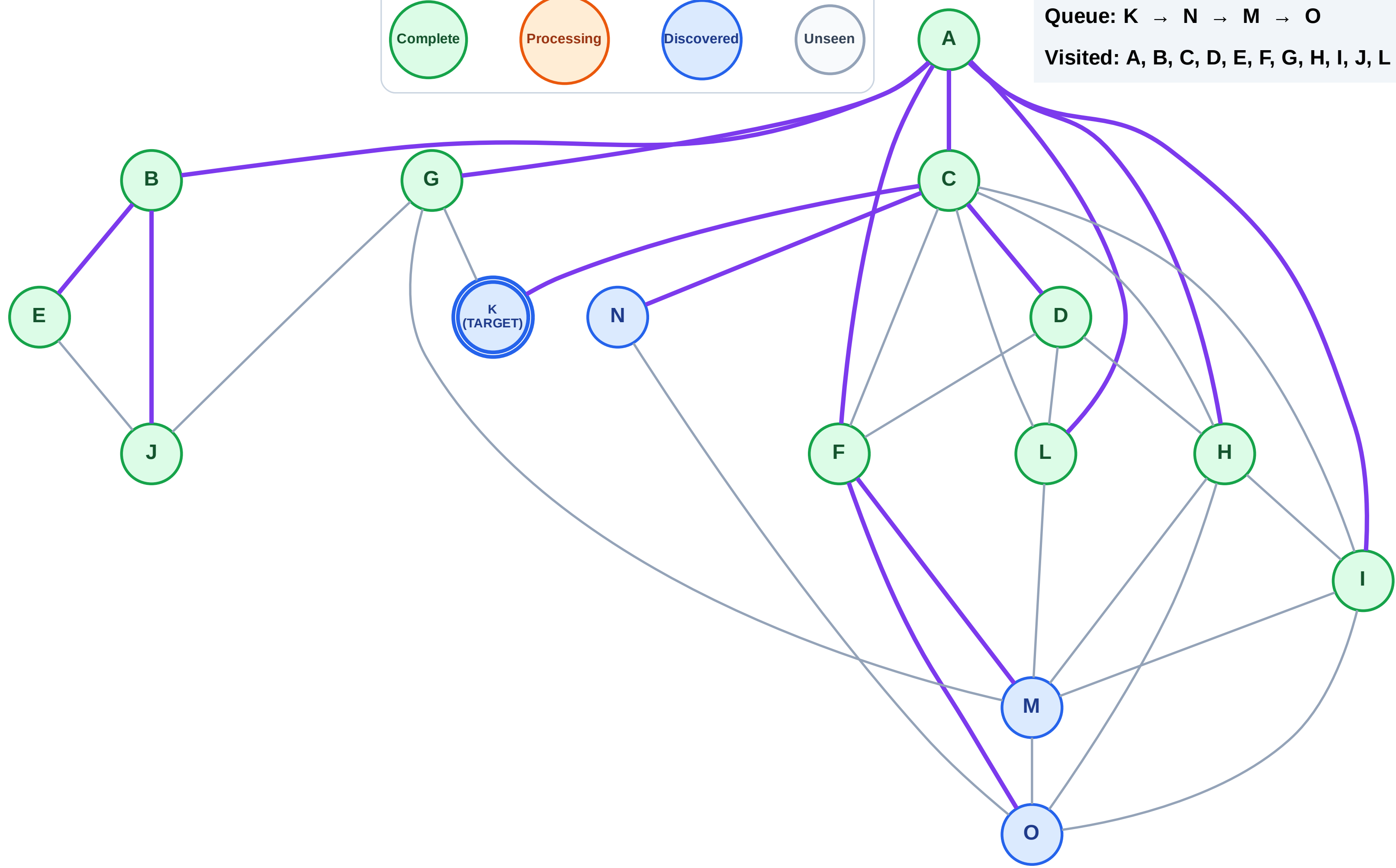
Discovered



Unseen

Queue: K → N → M → O

Visited: A, B, C, D, E, F, G, H, I, J, L



BFS Traversal

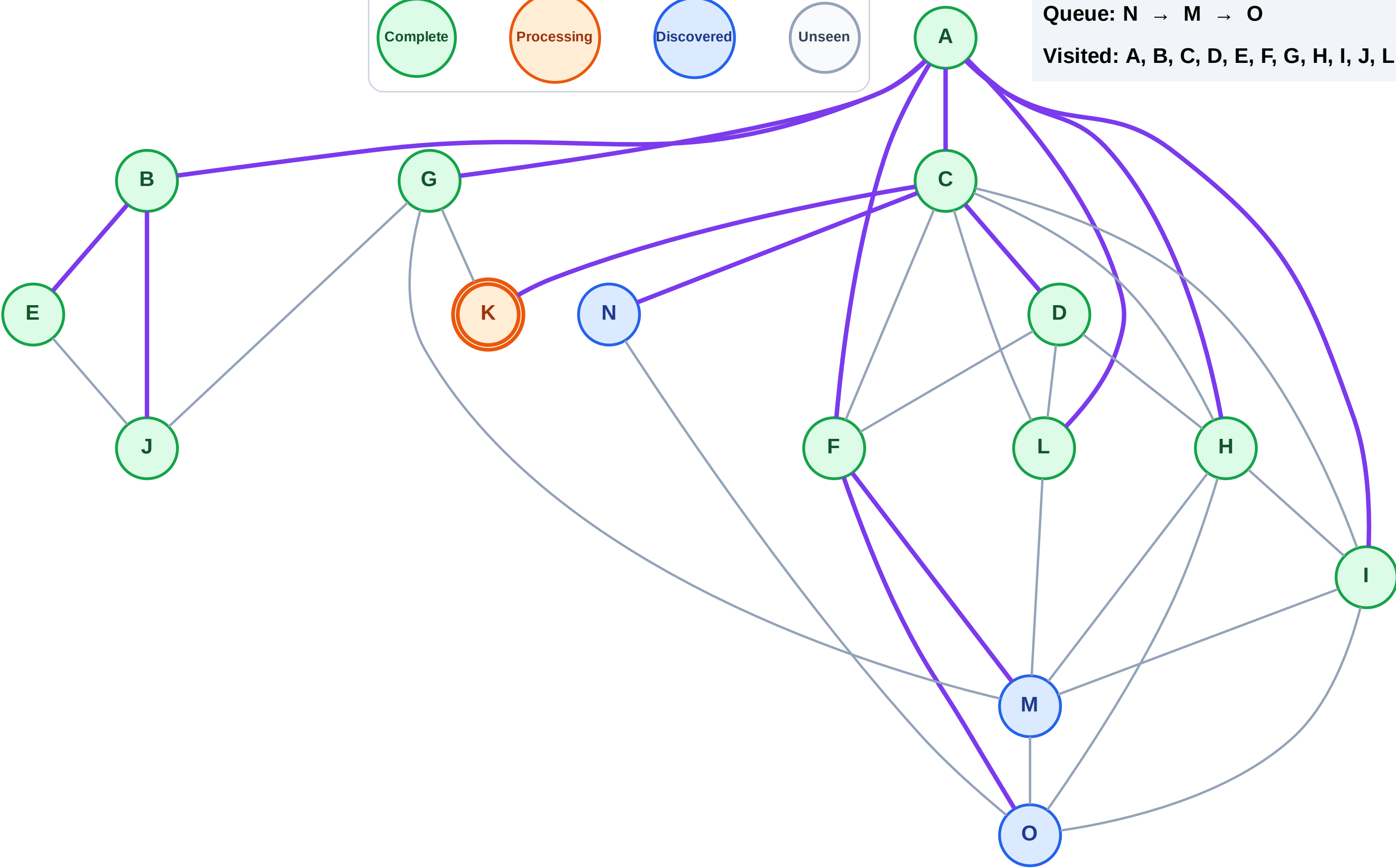
Step 24: Process node K

Legend



Queue: N → M → O

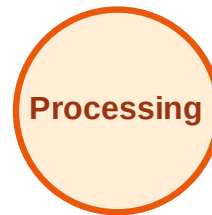
Visited: A, B, C, D, E, F, G, H, I, J, L



BFS Traversal

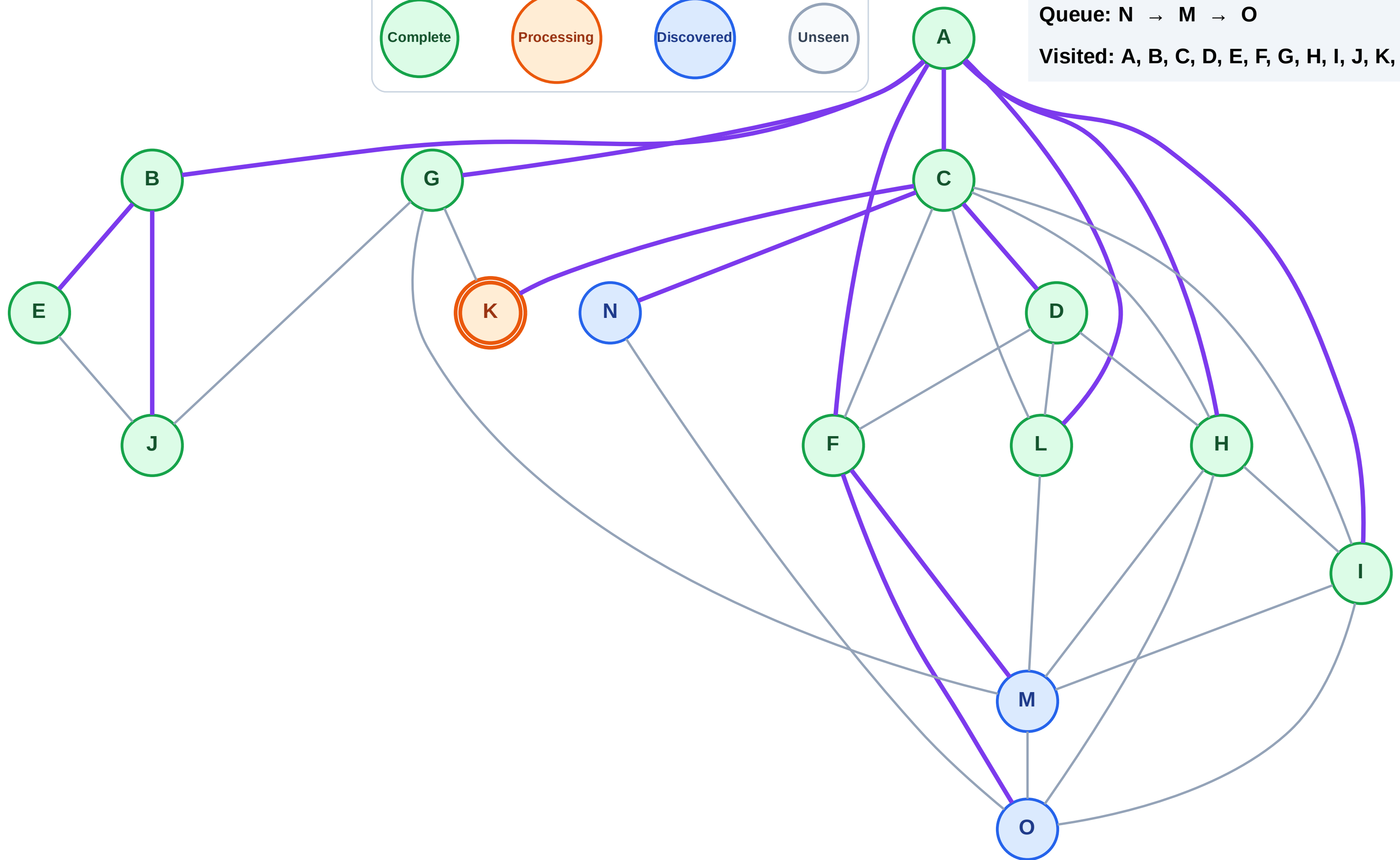
Step 25: TARGET NODE K REACHED! Path (2 hops): A → C → K

Legend



Queue: N → M → O

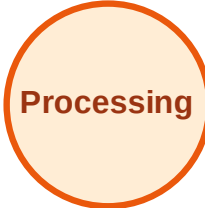
Visited: A, B, C, D, E, F, G, H, I, J, K, L



DFS Traversal

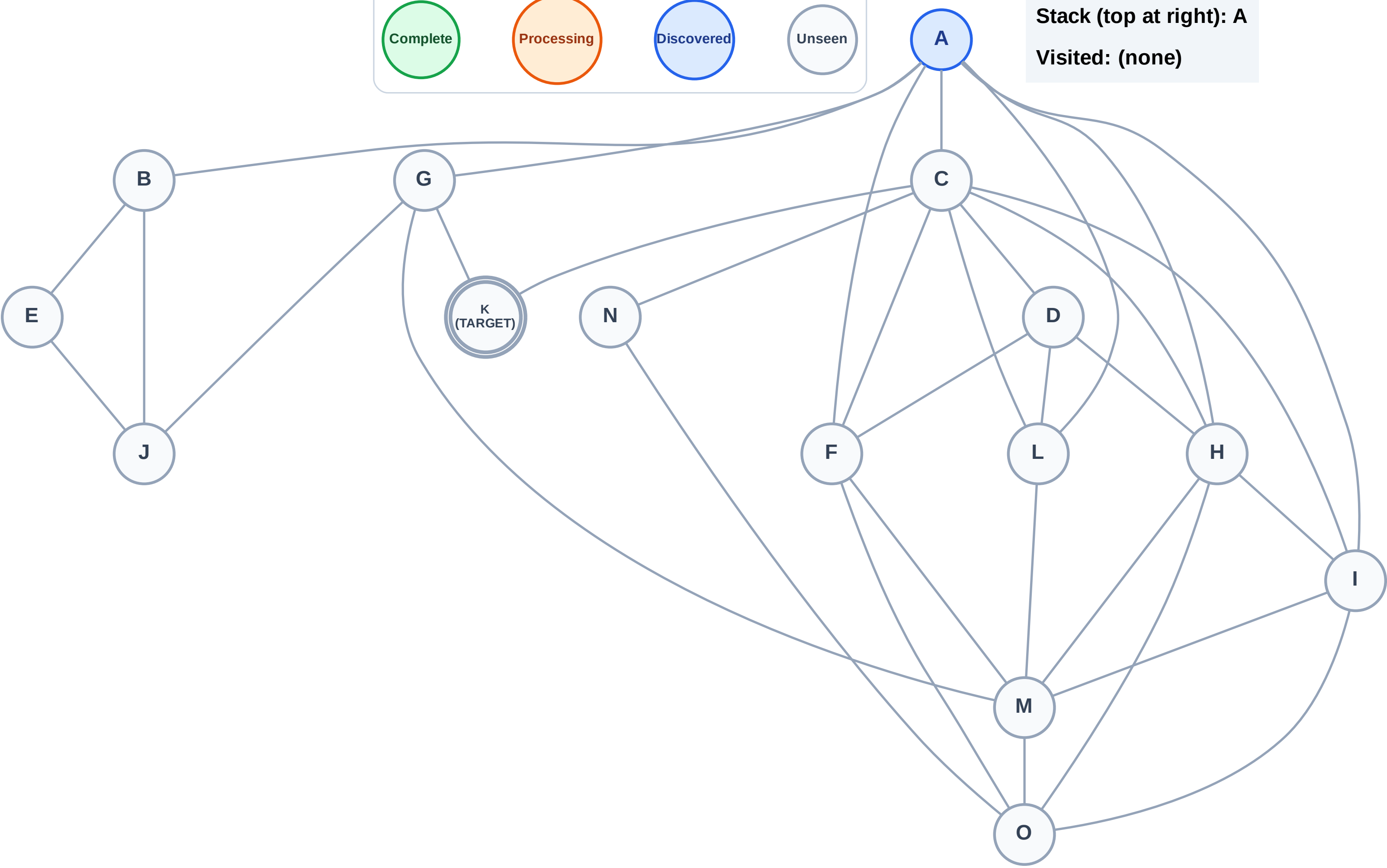
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

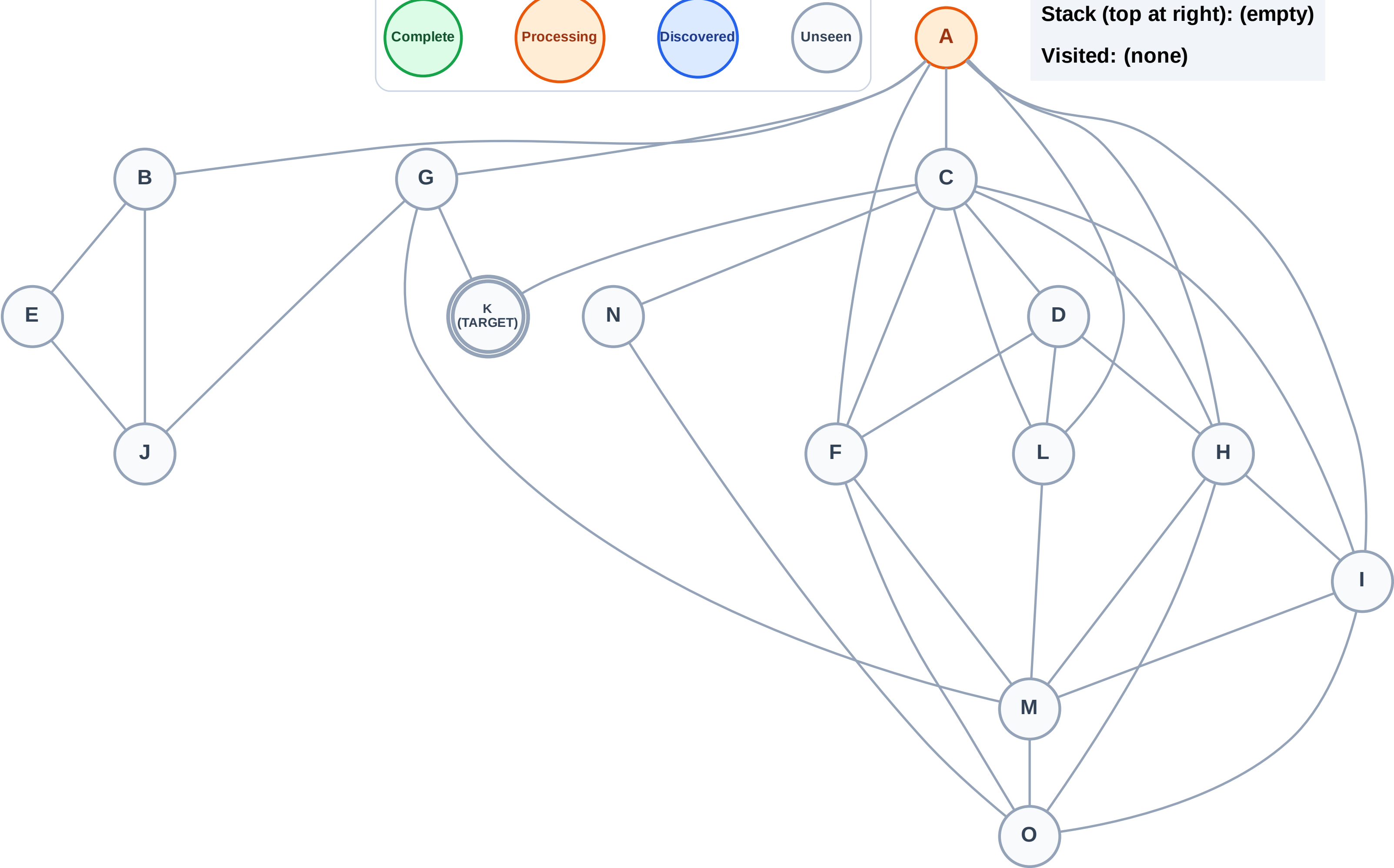
Step 2: Process node A

Legend



Stack (top at right): (empty)

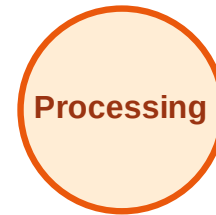
Visited: (none)



DFS Traversal

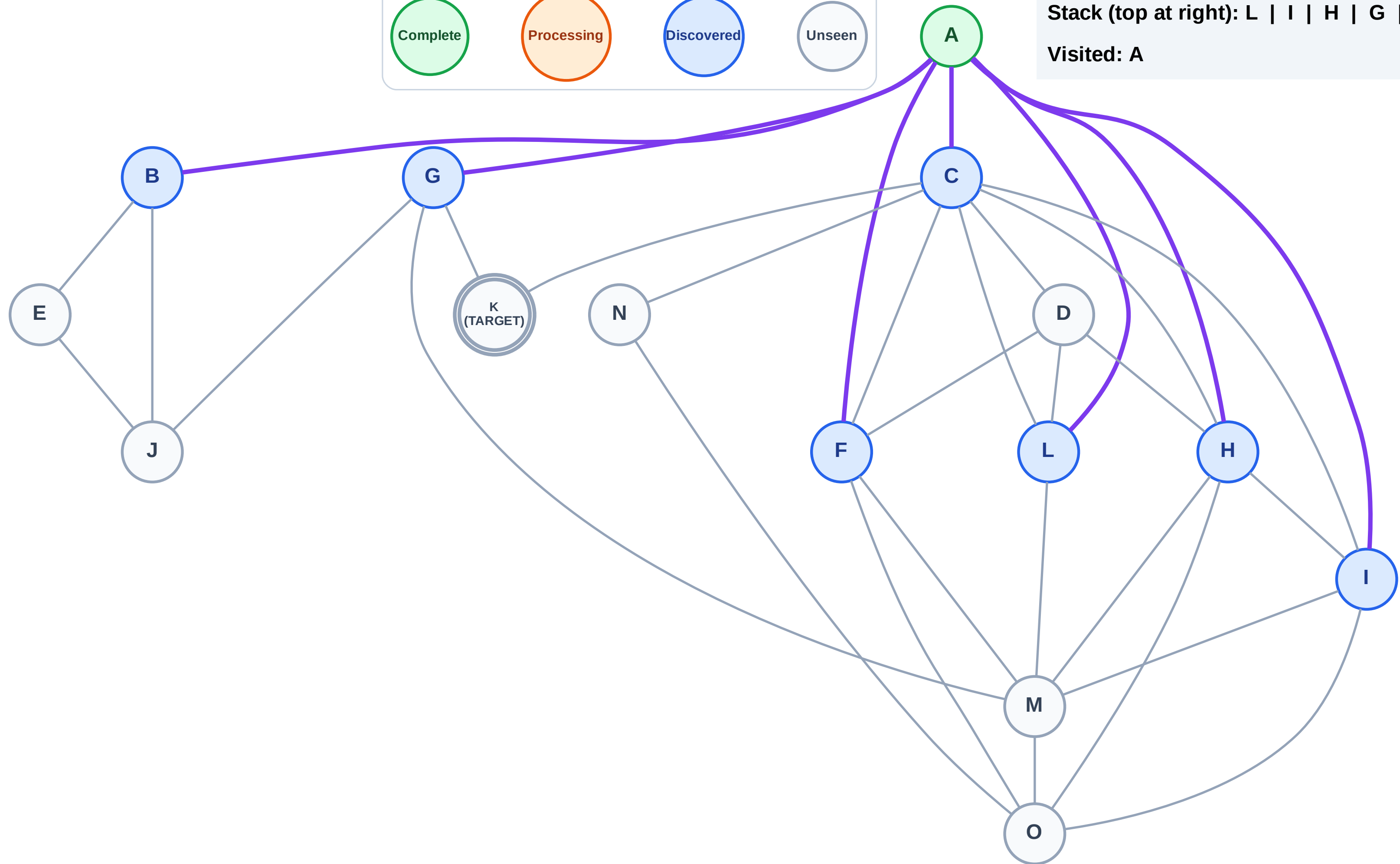
Step 3: Discover L, I, H, G, F, C, B from A

Legend



Stack (top at right): L | I | H | G | F | C | B

Visited: A



DFS Traversal

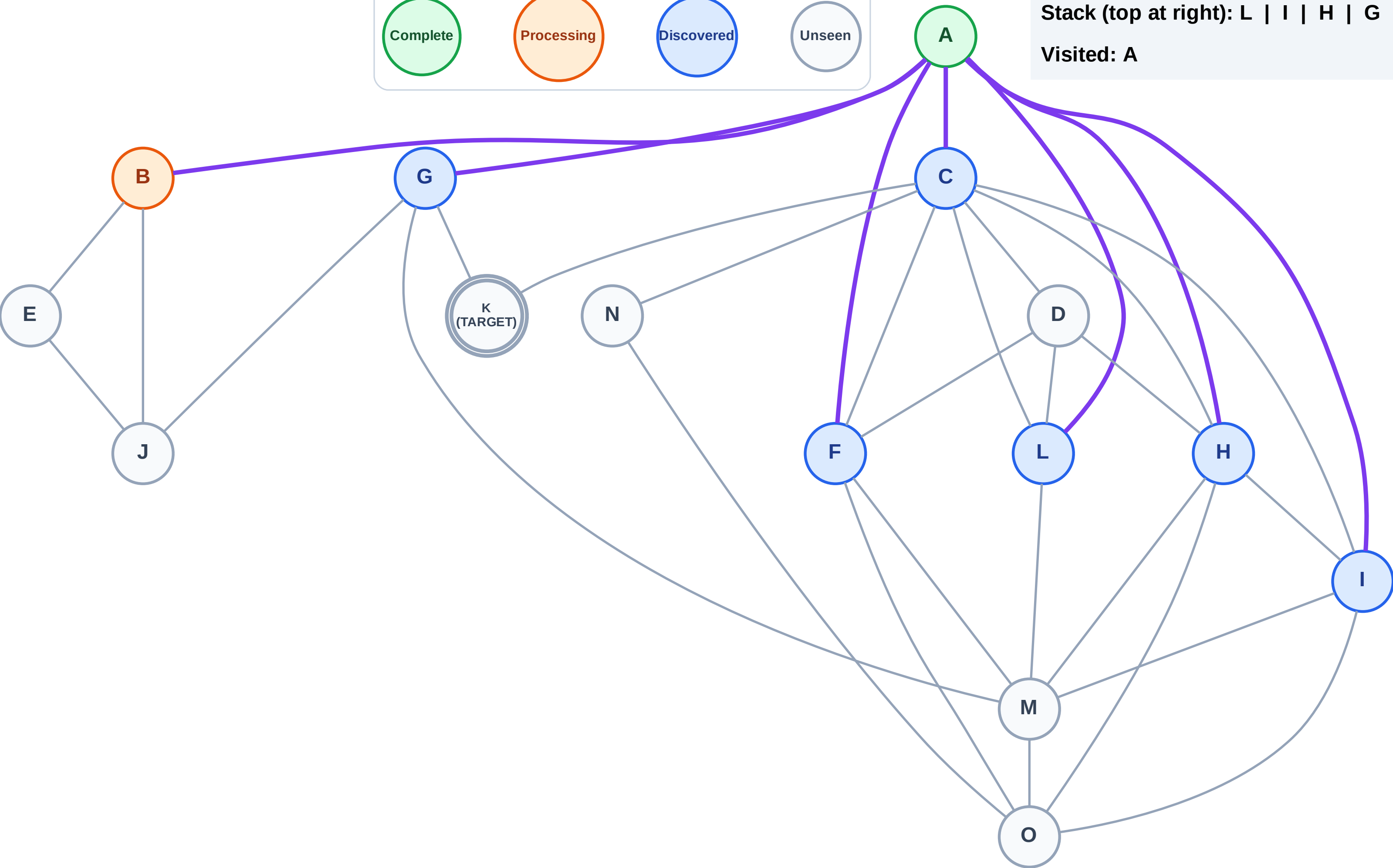
Step 4: Process node B

Legend



Stack (top at right): L | I | H | G | F | C

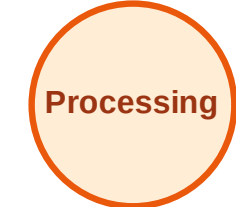
Visited: A



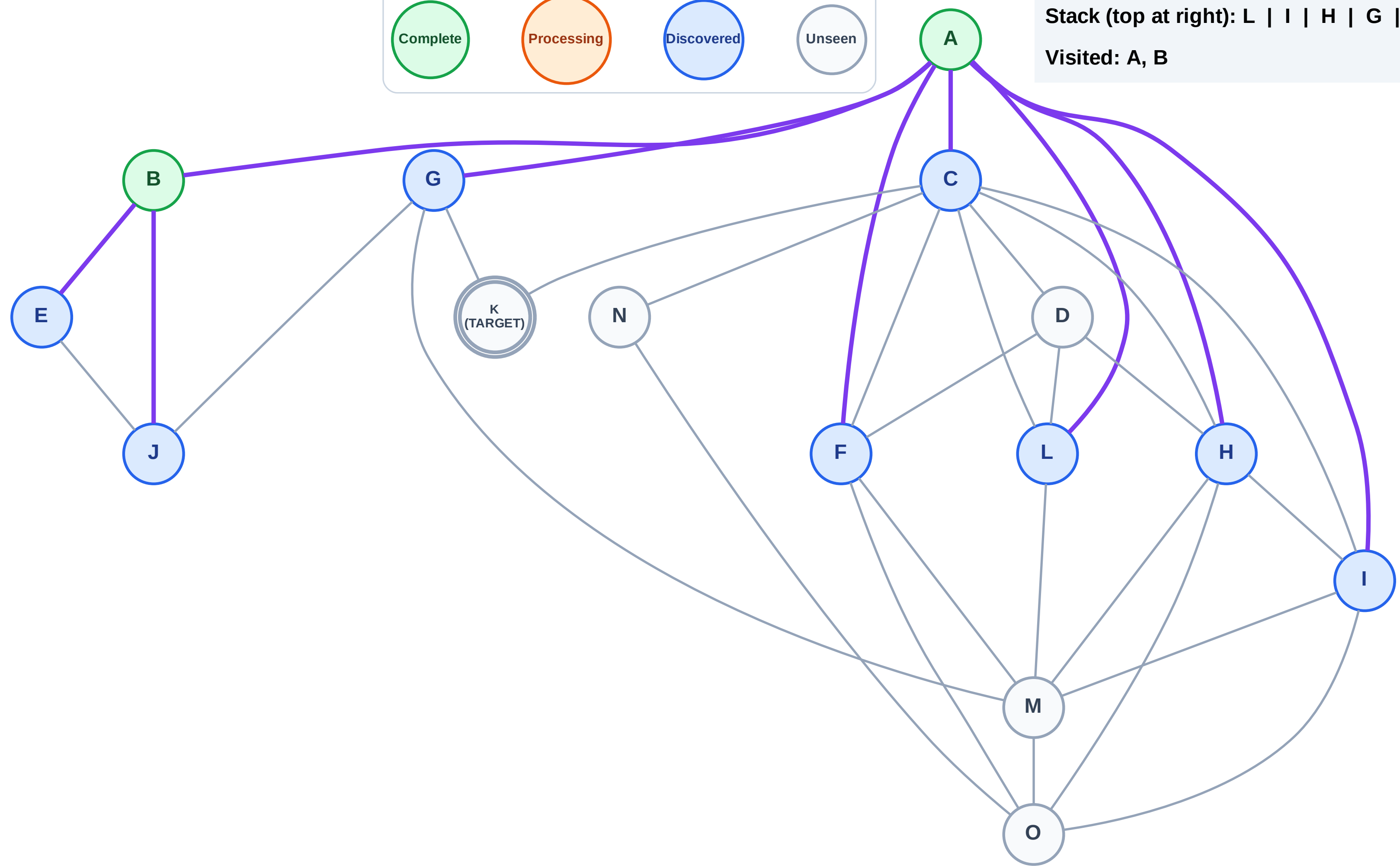
DFS Traversal

Step 5: Discover J, E from B

Legend



Stack (top at right): L | I | H | G | F | C | J | E
Visited: A, B



DFS Traversal

Step 6: Process node E

Legend



Complete



Processing



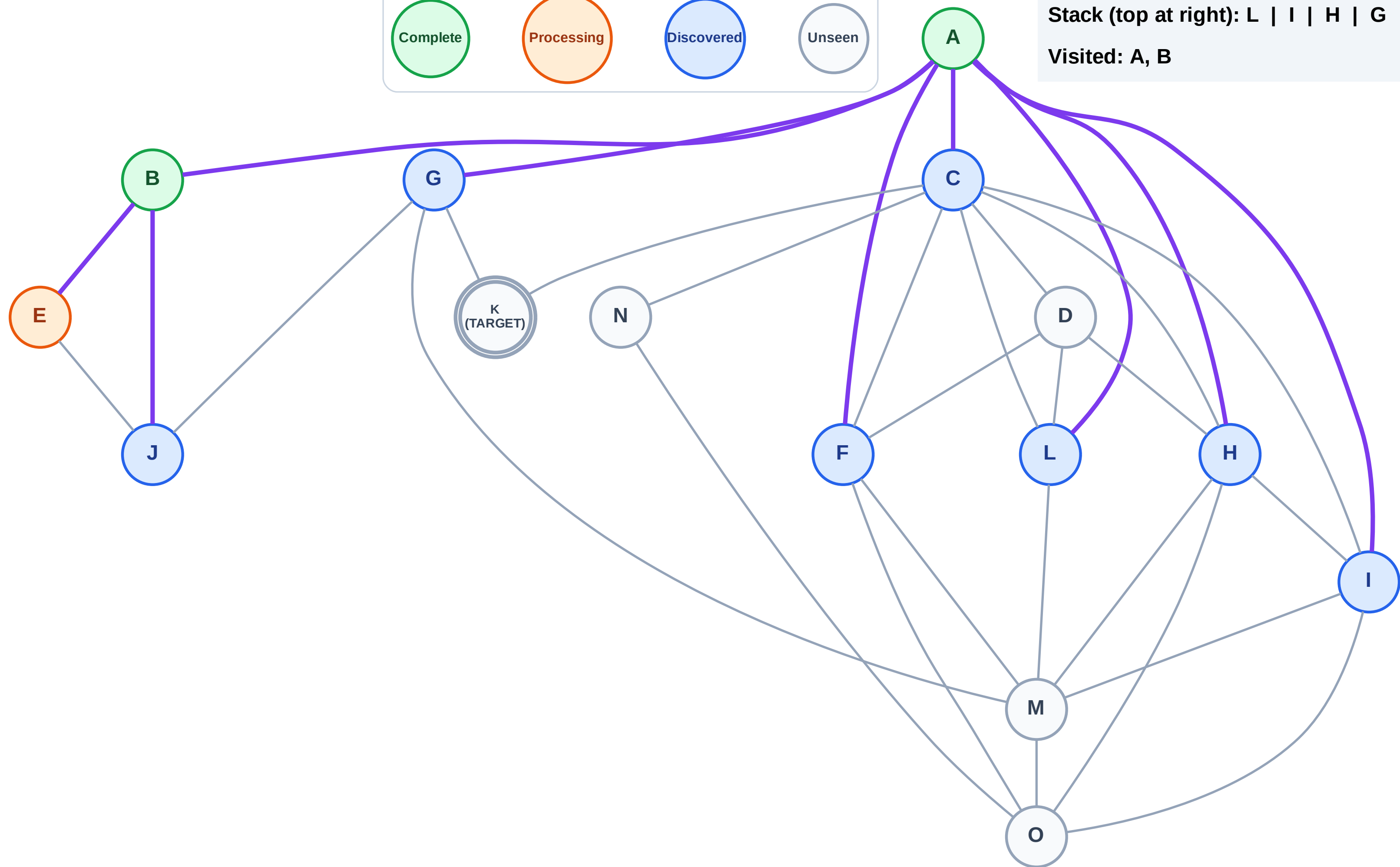
Discovered



Unseen

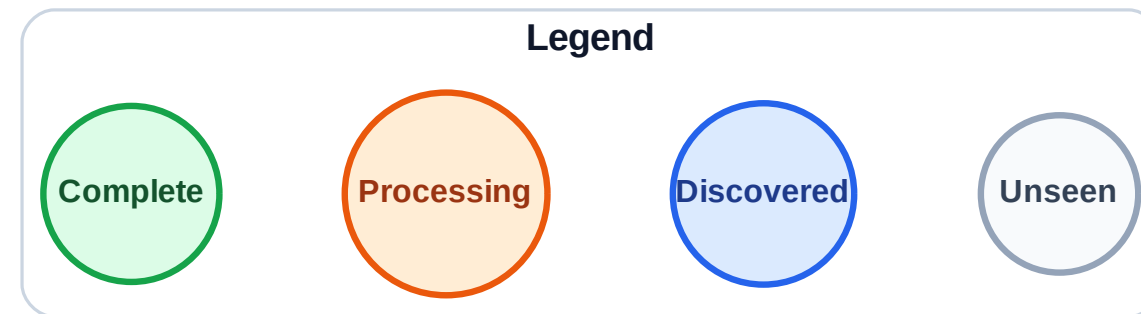
Stack (top at right): L | I | H | G | F | C | J

Visited: A, B



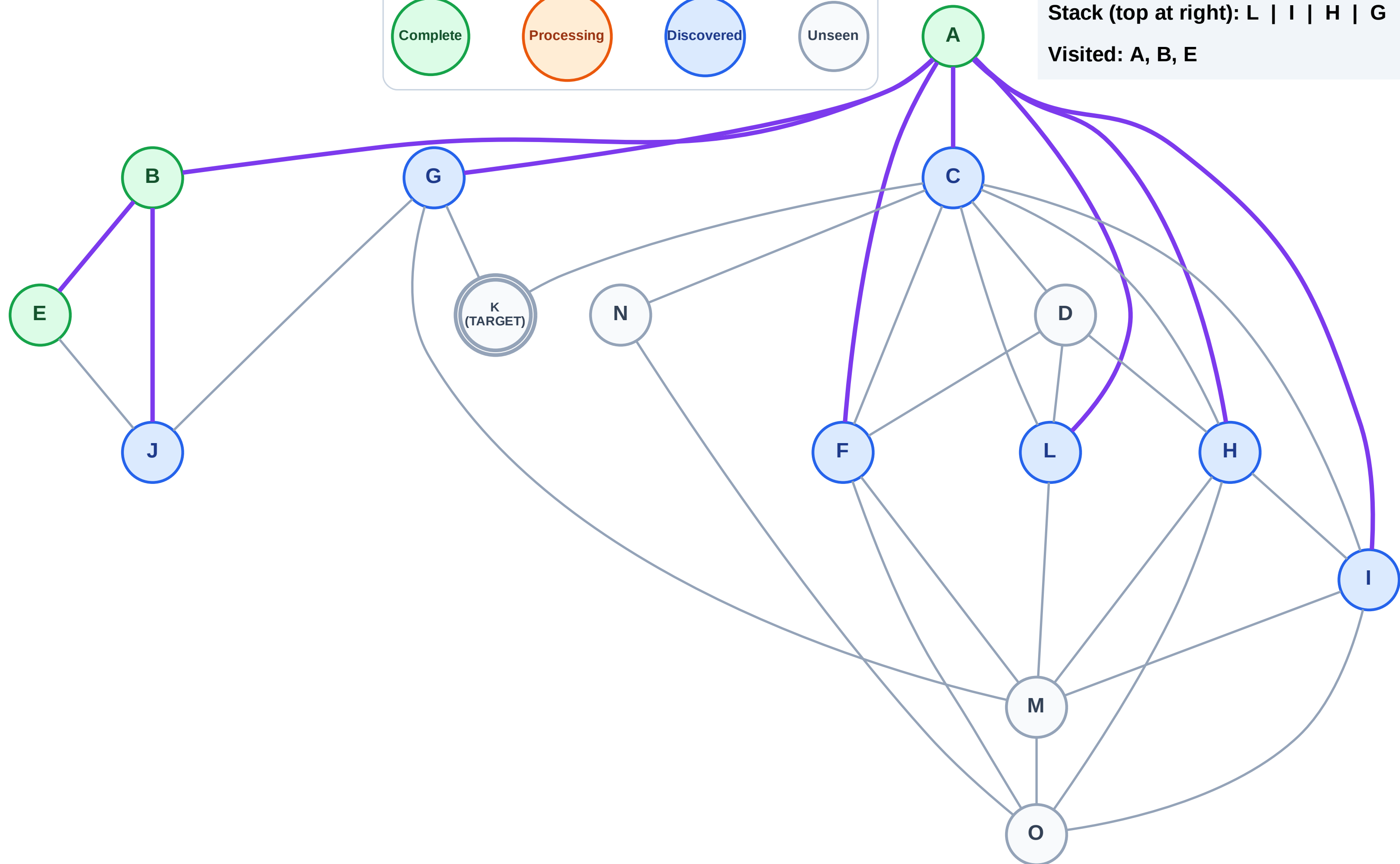
Step 7: No new neighbors from E

Step 7: No new neighbors from E



Stack (top at right): L | I | H | G | F | C | J

Visited: A, B, E



DFS Traversal

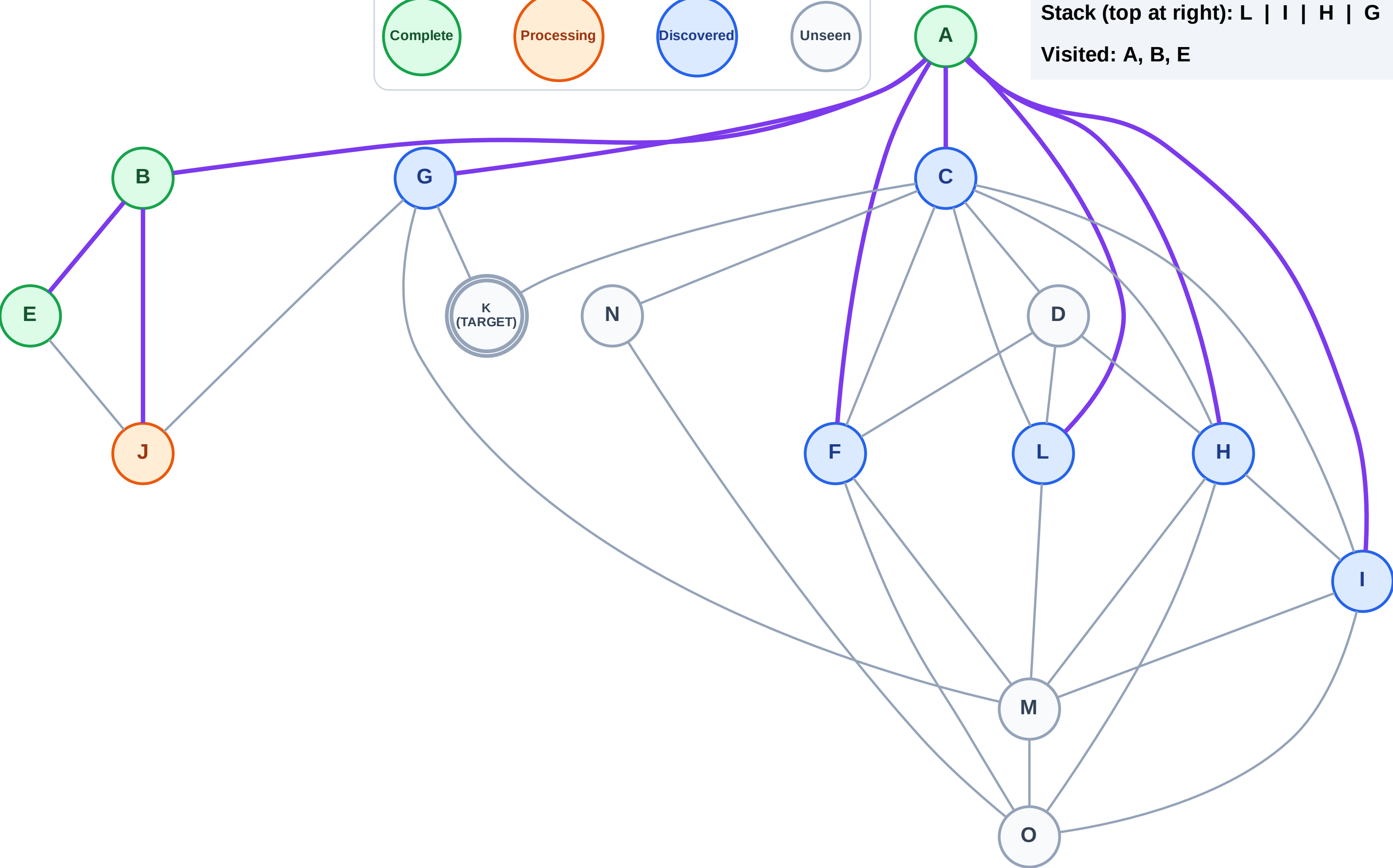
Step 8: Process node J

Legend



Stack (top at right): L | I | H | G | F | C

Visited: A, B, E



DFS Traversal

Step 9: No new neighbors from J

Legend



Complete



Processing

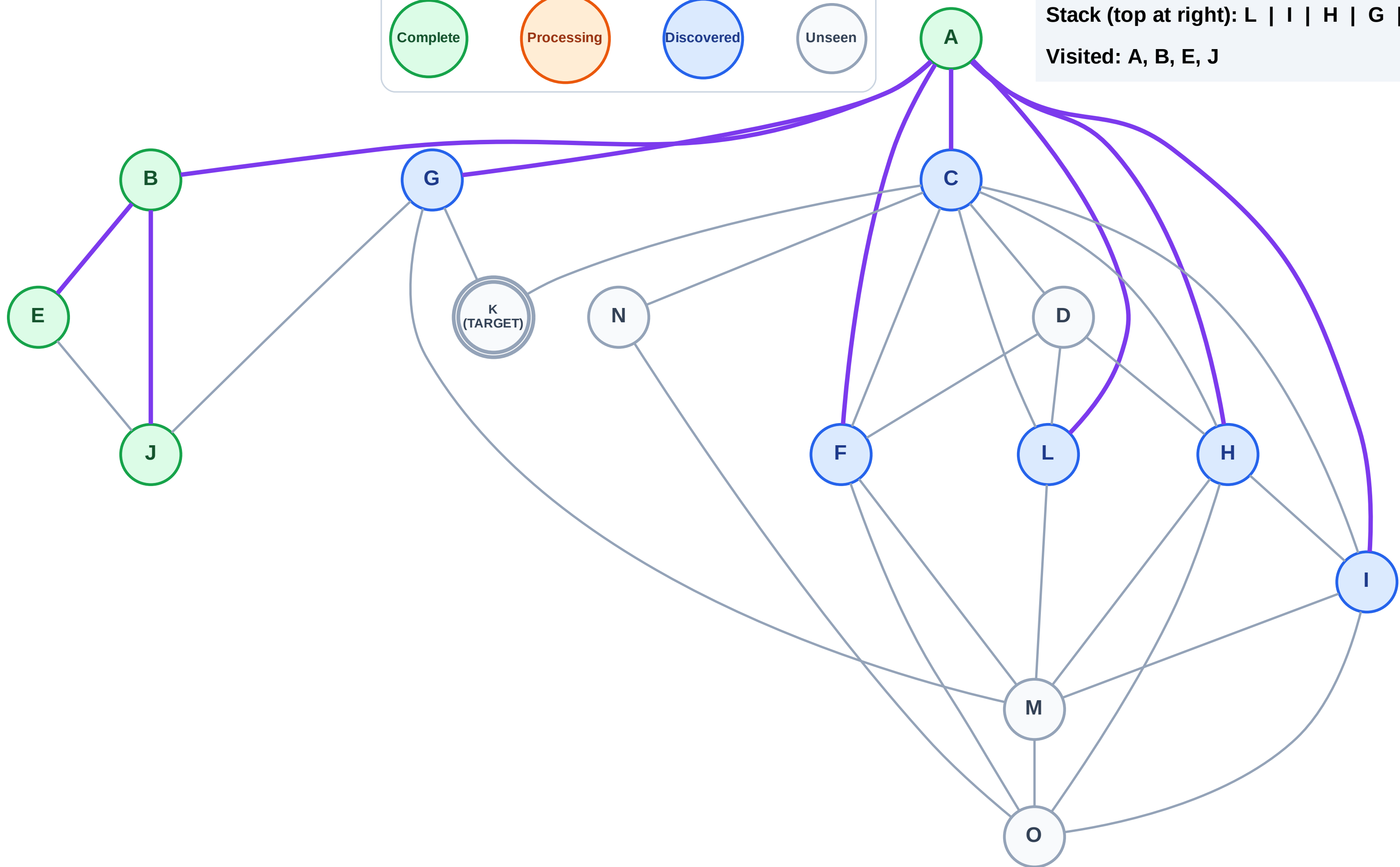


Discovered



Unseen

Stack (top at right): L | I | H | G | F | C
Visited: A, B, E, J



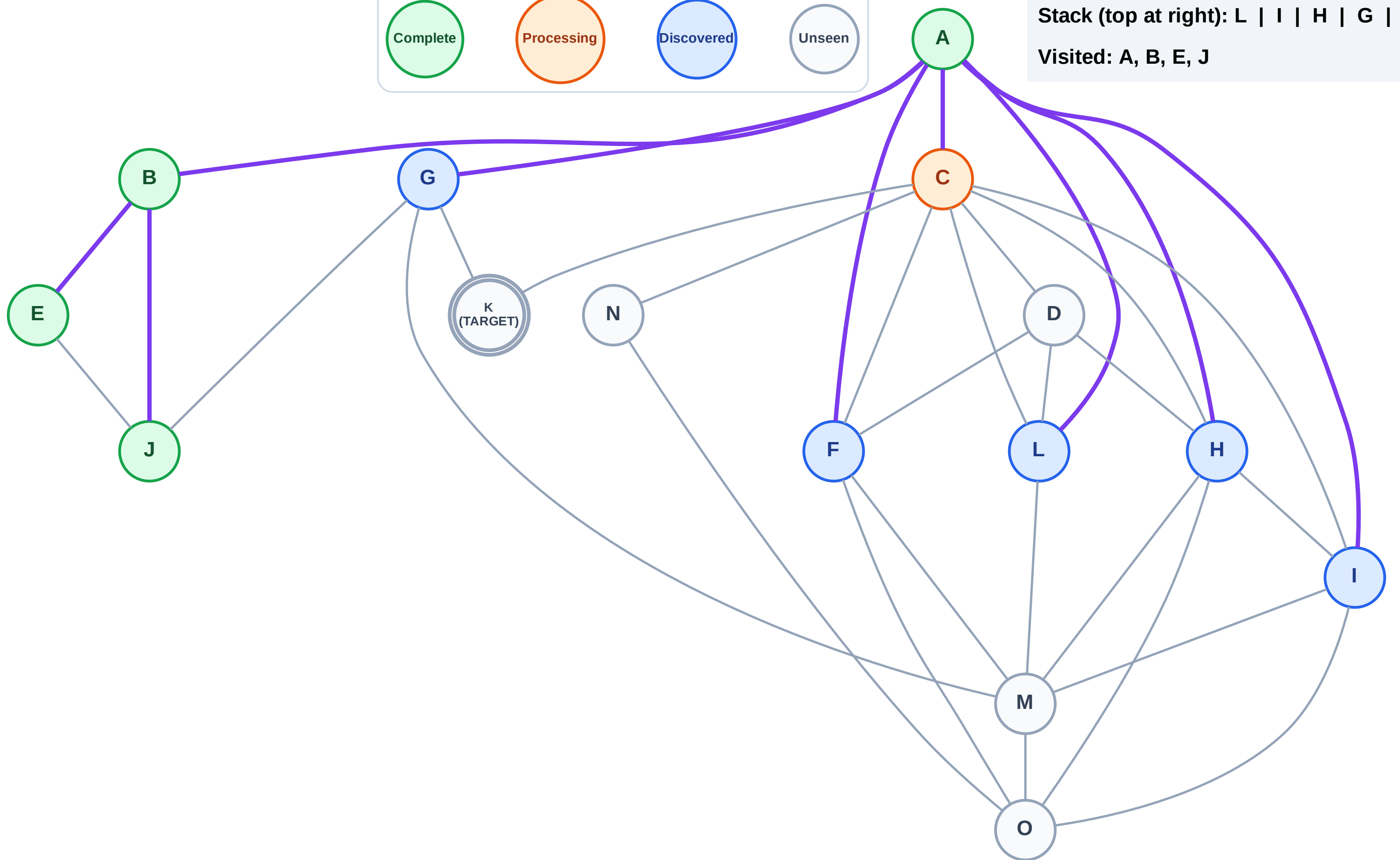
DFS Traversal

Step 10: Process node C

Legend



Stack (top at right): L | I | H | G | F
Visited: A, B, E, J



DFS Traversal

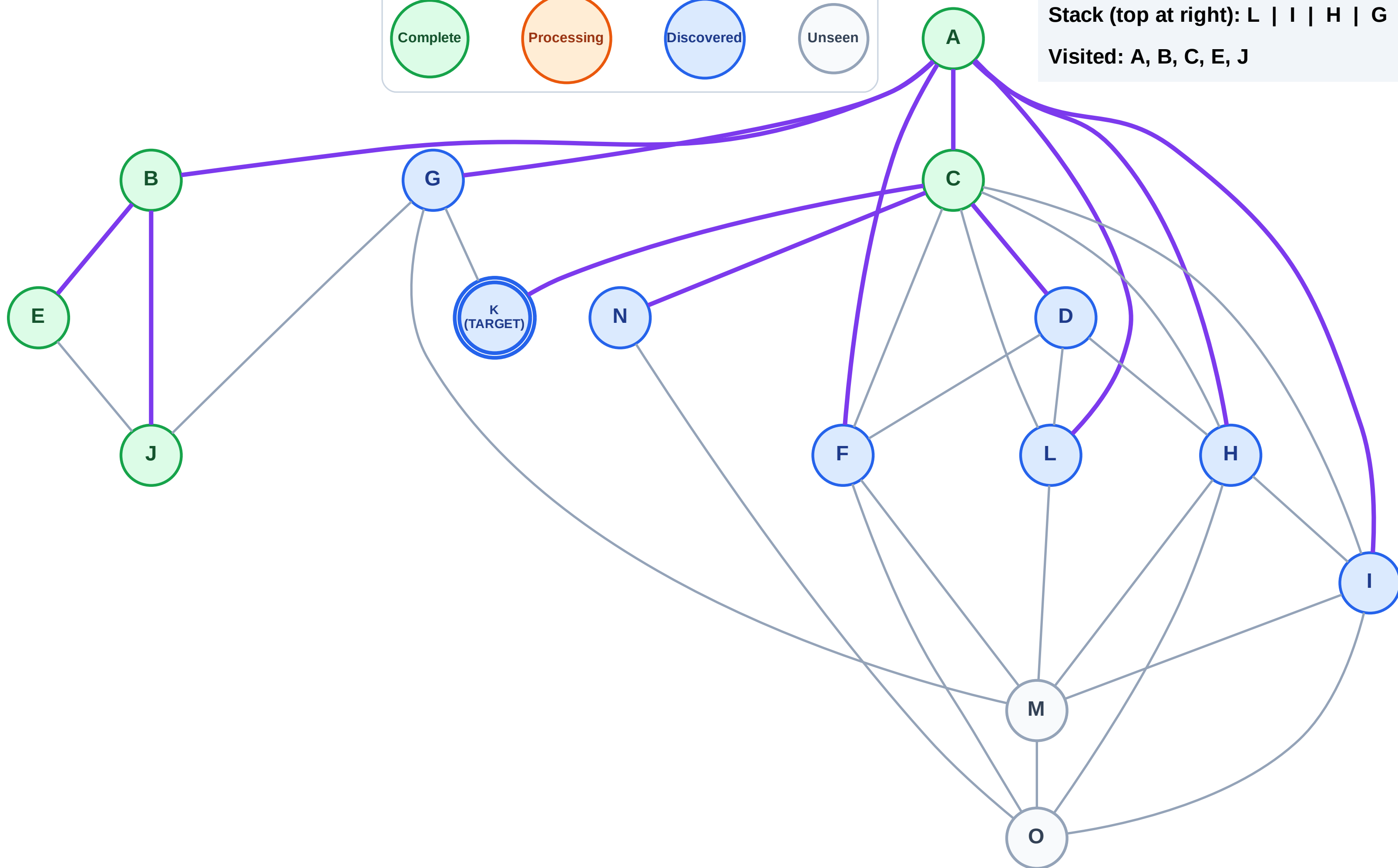
Step 11: Discover N, K, D from C

Legend



Stack (top at right): L | I | H | G | F | N | K | D

Visited: A, B, C, E, J



DFS Traversal

Step 12: Process node D

Legend

Complete

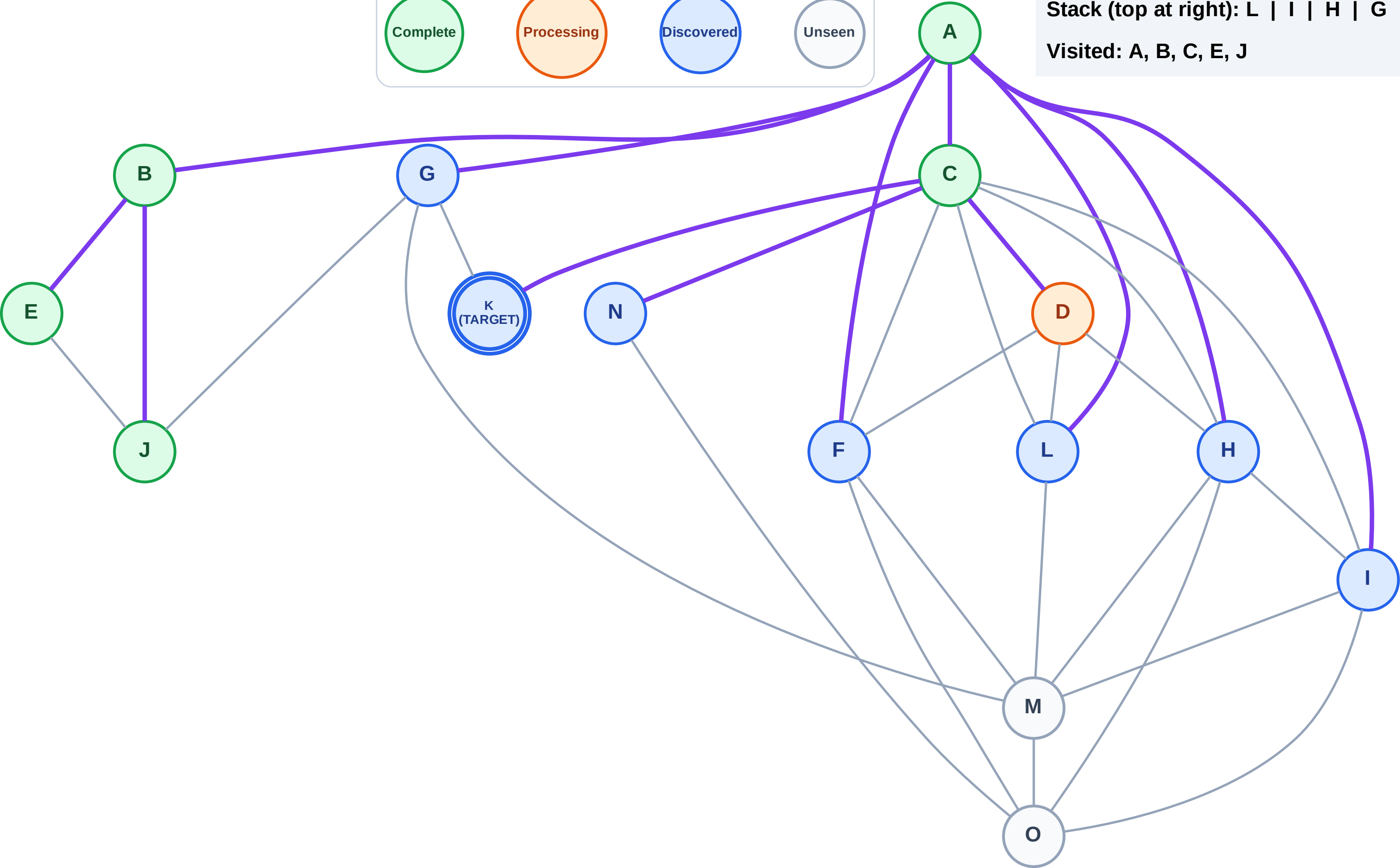
Processing

Discovered

Unseen

Stack (top at right): L | I | H | G | F | N | K

Visited: A, B, C, E, J



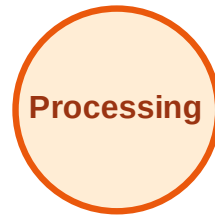
DFS Traversal

Step 13: No new neighbors from D

Legend



Complete



Processing



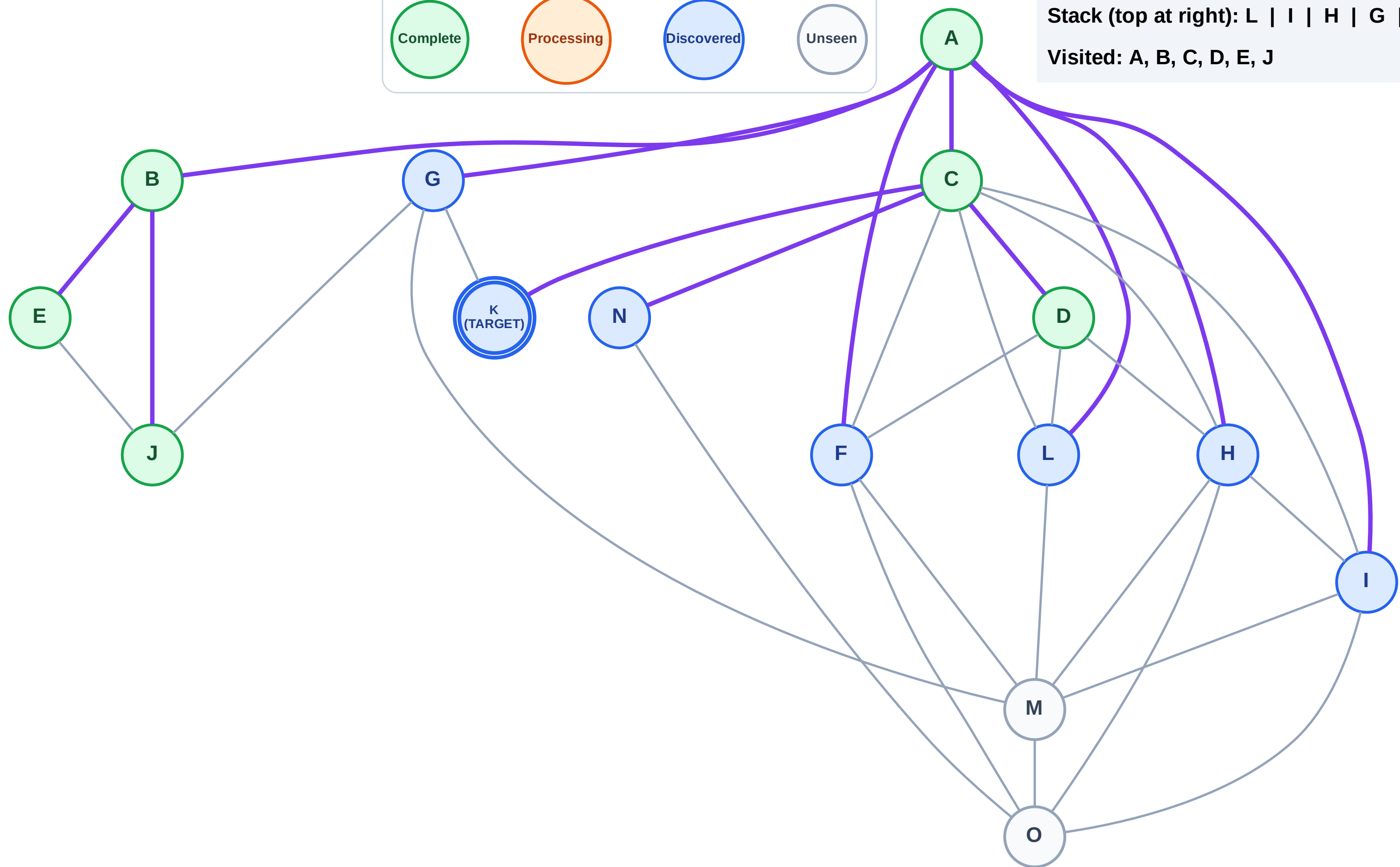
Discovered



Unseen

Stack (top at right): L | I | H | G | F | N | K

Visited: A, B, C, D, E, J



Step 14: Process node K

Step 14: Process node K

Legend

Complete

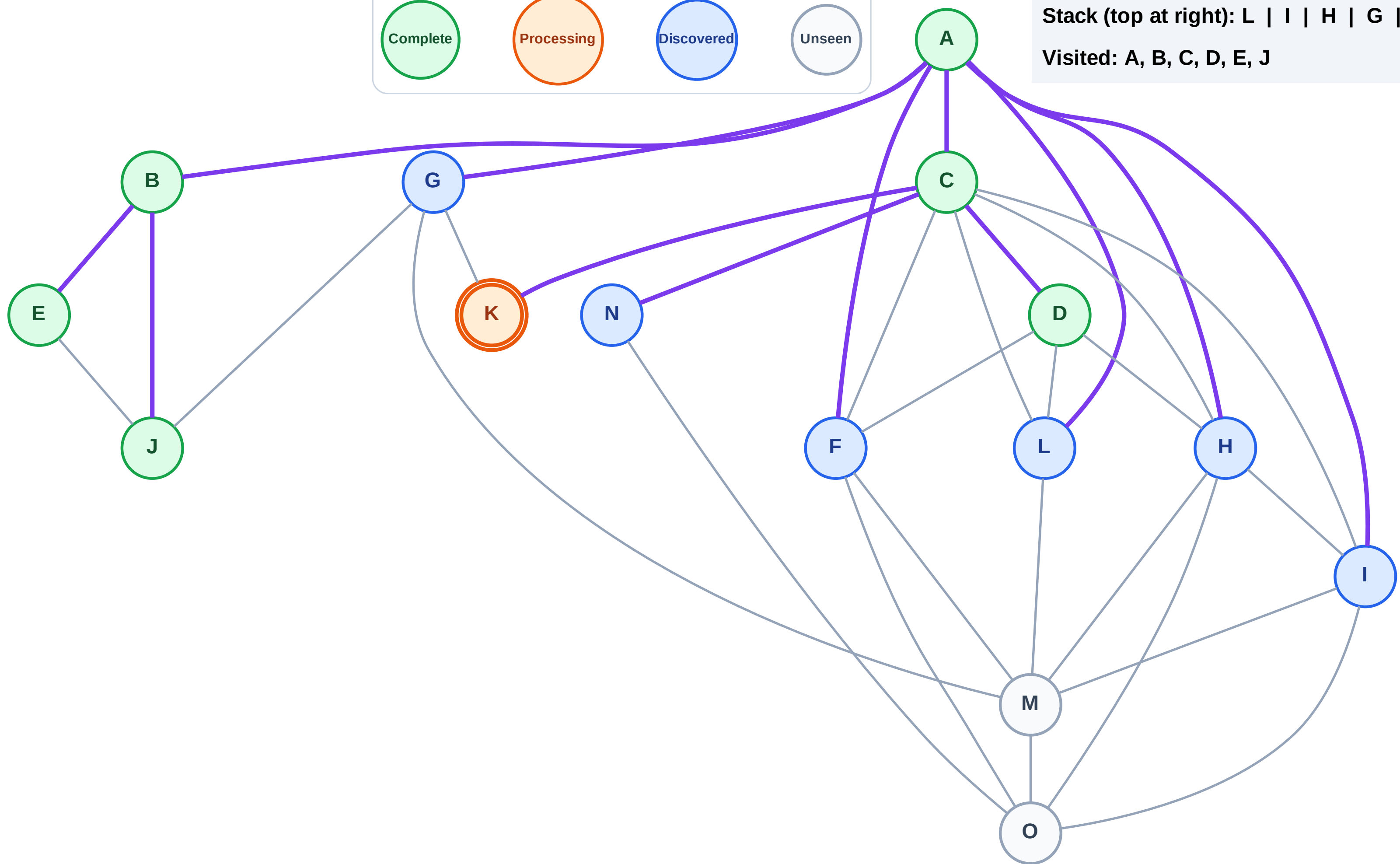
Processing

Discovered

Unseen

Stack (top at right): L | I | H | G | F | N

Visited: A, B, C, D, E, J



DFS Traversal

Step 15: TARGET NODE K REACHED! Path (2 hops): A → C → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): L | I | H | G | F | N

Visited: A, B, C, D, E, J, K

